STAT C205A: Probability Theory

By William Nguyen

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8/27/26 Lecture 1: Measure Theory

Classical probability is a good system, but it is not sufficient for finding the probability of infinite events, such as the probability that it takes an even number of coin flips before getting your first head. This is because we only have finite additivity, so what if we allow countable additivity?

Countable Additivity: If $B = \cup_{i=1}^{\infty} B_i$ and the B_i are pairwise disjoint, then say

$$P(B) = \sum_{i=1}^{\infty} P(B_i)$$

However, we run into problems. It turns out sets that we cannot naturally assign a probability to start to exist. Here is an example.

Example 1.1: Vitali Sets

Let our set of outcomes be S^1, the circle with radius 1 and circumference 2π. Suppose there is a probability measure P on $\mathcal{P}(S^1)$ that is rotationally invariant; that is, $P(B) = P(B + x)$. Now, we will construct a set that we cannot define a probability to.

Call $\omega', \omega'' \in S^1$ part of the same family if we can write $\omega' = \omega'' + n$ for some $n \in \mathbb{Z}$. Being part of a family is an equivalence relation, so consider the equivalence classes of this relation, and use the AoC to chose a representative from each family. Let $A \subseteq S^1$ be defined as follows

$$A = \{\omega \in S^1 \ : \ \omega \text{ is a representative}\}$$

Now, what is $P(A)$?

First, note that $(A + n) \cap A = \emptyset$ for $n \in \mathbb{Z}$, and that $\cup_{n\in\mathbb{Z}}(A + n) = S^1$. Because of rotational invariance, we see $P(A) = P(A + n)$, but we see

$$1 = P\left(\bigcup_{n\in\mathbb{Z}} (A + n)\right) = \sum_{n\in\mathbb{Z}} P(A + n) = \sum_{n\in\mathbb{Z}} P(A)$$

However, since $\mathbb{Z}$ is countable, there is no way of assigning a value to $P(A)$ that allows this to be true.

Thus, we must restrict our probability measuring functions to specific collections of subsets, which motivates the following definition.

Definition 1.1: σ-algebras / σ-fields

Let Ω be a set. A *σ-algebra* or *σ-field* in Ω is a collection $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ such that, if $A_i \in \mathcal{A}$ for $i = 1, 2, \ldots$, then A_i^c, $\cup_k A_k$, and $\cap_k A_k \in \mathcal{A}$. Also, $\Omega \in \mathcal{A}$ and $\emptyset \in \mathcal{A}$.

Fun Fact

A fun fact is that the "σ" in σ-algebra and σ-field stands for "countable", so whenever you see σ in this context, assume countability appears somewhere.

Simple examples of a σ-algebra is $\mathcal{A} = \{\emptyset, \Omega\}$ and $\mathcal{A} = \mathcal{P}(\Omega)$. Here are some more definitions that we wll soon use.

Definition 1.2: Measurable Spaces

A *measurable space* is a pair $(\Omega, \mathcal{A})$ where $\mathcal{A}$ is a σ-algebra.

Now, it can be checked quite easily that if A_λ is a σ-algebra for each $\lambda \in I$, then $\cap_\lambda A_\lambda$ is a σ-algebra. This allows us to make the following definition.

Definition 1.3: σ-algebra Generated by a Set

Given a subset $\mathcal{C} \subseteq \mathcal{P}(\Omega)$, $\sigma(\mathcal{C})$ is the intersection of all σ-algebras containing $\mathcal{C}$. We call $\sigma(\mathcal{C})$ the *σ-algebra generated by $\mathcal{C}$*.

Definition 1.4: Borel σ-algebras

Let S be a metric/topological space. If G is the set of all open sets in S, let $\mathcal{B}(S) := \sigma(G)$. We call $\mathcal{B}(S)$ the *Borel σ-algebra* on S.

Definition 1.5: π-systems

A class $\mathcal{C} \subseteq \mathcal{P}(\Omega)$ is called a *π-system* if it is closed under finite intersections.

Definition 1.6: λ-systems

A class $\mathcal{D} \subseteq \mathcal{P}(\Omega)$ is called a *λ-system* if it contains Ω and is closed under proper differences and increasing limits. That is, if $A, B \in \mathcal{D}$ with $B \subseteq A$, then $A \backslash B \in \mathcal{D}$, and if $A_1, A_2, \ldots \in \mathcal{D}$ with $A_1 \uparrow A$ (that is; $A_1 \subseteq A_2 \subseteq \cdots \subseteq A = \cup_i A_i$), then $A \in \mathcal{D}$.

9/1/26 Lecture 2: Topology

Theorem 2.1: Dynkin's π-λ Theorem

If $\mathcal{C}$ is a π-system and $\mathcal{D}$ is a λ-system such that $\mathcal{C} \subseteq \mathcal{D}$, then $\sigma(\mathcal{C}) \subseteq \mathcal{D}$.

Proof: First, note that as set S is a σ-algebra if, and only if, it is both a π and λ system. Clearly, we have the $\Rightarrow$ direction, and for the $\Leftarrow$ direction, note that if S is a π and λ system, it contains $\Omega, \emptyset$, complements, and countable unions (De Morgan's + S is a π-system $\implies$ closed under finite unions + increasing limits $\implies$ closed under countable unions).

Now, let $\lambda(\mathcal{C})$ be the smallest λ-system containing $\mathcal{C}$ (it is clear that the intersection of λ-systems is a λ-system so this definition makes sense). We just need to show that $\lambda(\mathcal{C})$ is a σ-algebra, which is to show that it is closed under intersections (making it a π-system)

Let $A, B \in \lambda(\mathcal{C})$. If both $A, B \in \mathcal{C}$, then clearly $A \cap B \in \lambda(C)$ since $A \cap B \in \mathcal{C}$. Now, if $B \in \mathcal{C}$, then consider the set $\mathcal{A}_B = \{A \subseteq \Omega : A \cap B \in \lambda(\mathcal{C})\}$. Note that $\mathcal{A}_B$ contains $\mathcal{C}$. It also contains Ω. Now, if $F \subseteq G$ with $F, G \in \mathcal{A}_B$, since $F \cap B, G \cap B \in \lambda(\mathcal{C})$ we see that $F \cap B \subseteq G \cap B$, so $(G \cap B) \backslash (F \cap B) = (G \backslash F) \cap B \in \lambda(\mathcal{C})$, so $G \backslash F \in \mathcal{A}_B$.

Finally, let $A_i \uparrow A$ with $A_i \in \mathcal{A}_B$. Then, as $A_i \cap B \in \lambda(\mathcal{C})$ and $A_i \cap B \uparrow A \cap B$, we see that $A \cap B \in \lambda(\mathcal{C})$, so $A \in \mathcal{A}_B$. Thus, $\mathcal{A}_B$ is a λ-system containing $\mathcal{C}$, so $\lambda(\mathcal{C}) \subseteq \mathcal{A}_B$. Thus, we see that if $A \in \lambda\mathcal{C}$ and $B \in \mathcal{C}$, then $A \cap B \in \lambda(\mathcal{C})$.

Finally, if $A \in \lambda(\mathcal{C})$, consider the set $\mathcal{A}'_A = \{B \subseteq \Omega : A \cap B \in \lambda(\mathcal{C})\}$. By the discussion above, we see that $\mathcal{C}$ is contained in $\mathcal{A}'_A$, and by the same argument as above, $\mathcal{A}'_A$ is a λ-system. Thus, $\lambda(\mathcal{A}) \subseteq \mathcal{A}'_A$, which implies that $A \cap B \in \lambda(\mathcal{A})$.

Then, as $\sigma(\mathcal{C}) \subseteq \lambda(\mathcal{C}) \subseteq \mathcal{D}$, we have our desired inclusion.

■

This theorem is useless for now, but will soon prove itself useful! Now, some topology.

Definition 2.2: Base for a Topology

A *base* for a topological space S is a collection of open sets $\mathcal{C}$ such that every open set in S is a union of sets in $\mathcal{C}$. Equivalently, a collection $\mathcal{C}$ is a base if for all open $\mathcal{U}$ with $x \in \mathcal{U}$, there exists a $C \in \mathcal{C}$ such that $x \in C \subseteq \mathcal{U}$.

Definition 2.3: Second-Countable Spaces

A topological space S is *second-countable* if it has a countable base.

It can be shown fairly easily that, for metric spaces, second-countability is equivalent to separability, which means that the metric space has a countable dense subset.

Definition 2.4: Cartesian Product

Suppose for some index set T, we have sets X_t, $t \in T$. If $\sqcup_{t\in T} X_t$ is the disjoint union of these sets, then the *Cartesian product*, $\bigtimes_{t\in T} X_t$ is the set of maps f from T to $\sqcup_{t\in T} X_t$ such that $f(t) \in X_t$ for all $t \in T$.

For example, we see that $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ is the collection of functions $f : \{1, 2\} \to \mathbb{R} \sqcup \mathbb{R}$ such that $f(1) \in \mathbb{R}$ and $f(2) \in \mathbb{R}$.

Definition 2.5: Product Topology / Tychonoff Topology

Given topological spaces S_t, $t \in T$ for some index set T, the *product topology* (or *Tychonoff topology*) on $\bigtimes_{t\in T} S_t$ is such that open sets in $\bigtimes_{t\in T} S_t$ can be written as the union of sets of the form $\bigtimes_{t\in T} \mathcal{U}_t$ where $\mathcal{U}_t$ are open in S_t and $\mathcal{U}_t = S_t$ for all but finitely many $t \in T$.

In other words, the product topology are generated by sets of the form $\bigtimes_{t\in T} \mathcal{U}_t$ with the properties mentioned.

One thing a reader may be curious is why we require the finiteness of $\mathcal{U}_t \neq S_t$? The topology where we don't require that, called the *box topology* is still a topology, but it isn't as nice. For instance, the product of compact topological spaces is compact under the product topology, but not necessarily under the box topology.

Because of this, if Ω_t are topological spaces, then we always endow $\bigtimes_t \Omega_t$ with the product topology unless otherwise mentioned.

Lemma 2.6

If $S_1, S_2, \ldots$ are second-countable topological spaces, then so is $\bigtimes_{t=1}^{\infty} S_t$.

Proof: Let $\mathcal{C}_n$ be a countable base for each n. Now, let $\mathcal{C}$ consist of sets of the form $\bigtimes_{t=1}^{\infty} C_n$ where $C_n \in \mathcal{C}_n$ and $C_n = S_n$ for all but finitely many n. We see that $\mathcal{C}$ is countable.

Now, recall that the product topology on $\bigtimes_{t=1}^{\infty} S_t$ has base $\mathcal{U} = \bigtimes_{t=1}^{\infty} \mathcal{U}_t$ where $\mathcal{U}_t$ is open and $\mathcal{U}_t = S_t$ for all but finitely $\mathcal{U}_t$. Now, for $(x_1, x_2, \ldots) \in \mathcal{U}$, since we can find $C_i \in \mathcal{C}_i$ such that $x_i \in C_i \subseteq \mathcal{U}_i$, for the finitely many i with $\mathcal{U}_i \neq S_i$, we see that if $C = \bigtimes_{t=1}^{\infty} C_t$, we have $C \in \mathcal{C}$ and $x \in C \subseteq \mathcal{U}$. Thus, $\mathcal{C}$ is also a base for $\bigtimes_{t=1}^{\infty} S_t$, so it has a countable base.

■

Definition 2.7: Product σ-Field

Let Ω_t, $t \in T$ be measurable spaces, each with a σ-field $\mathcal{A}_t$. Then, for the Cartesian product, $\bigtimes_{t\in T} \Omega_t$, the *product σ-field* on $\bigtimes_{t\in T} \Omega_t$, denoted $\bigotimes_{t\in T} \mathcal{A}_t$ is the σ-field generated by the *cylinder sets* $A_t \times \bigtimes_{s\neq t} \Sigma_s$ for $A_t \in \mathcal{A}_t$.

Lemma 2.7

If $S_1, S_2, \dots$ are second-countable topological spaces, then

$$\mathcal{B}(S_1 \times S_2 \times \cdots) = \mathcal{B}(S_1) \otimes \mathcal{B}(S_2) \otimes \cdots$$

where $\mathcal{B}(X)$ gives the Borel σ-algebra on a topological space X.

We will not prove this and just move on. Now, recall that every point map $f : S \to T$ induces a set map $f^{-1} : \mathcal{P}(T) \to \mathcal{P}(S)$ by

$$f^{-1}(B) = \{s \in S \,:\, f(s) \in B\}$$

Note that f^{-1} commutes with complements, countable unions, and countable intersections; that is

$$f^{-1}(A^c) = (f^{-1}(A))^c, \quad f^{-1}\left(\bigcup_{i=1}^{\infty} A_i\right) = \bigcup_{i=1}^{\infty} f^{-1}(A_i), \quad \text{and} \quad f^{-1}\left(\bigcap_{i=1}^{\infty} A_i\right) = \bigcap_{i=1}^{\infty} f^{-1}(A_i)$$

so that motivates the following lemma.

Lemma 2.8

Let $f : S \to T$ where $(S, \mathcal{S})$ and $(T, \mathcal{T})$ are measurable spaces. Then,

1. $\mathcal{S}' = f^{-1}(\mathcal{T})$ is a σ-field on S.
2. $\mathcal{T}' = \{B \subseteq T \,:\, f^{-1}(B) \in \mathcal{S}\}$ is a σ-field on T.

9/3/26 Lecture 3: Digression on Product Topology

Question: Why the product topology instead of the box topology?

First, note that we can define the product topology as the one generated by sets of the form $(a_1, b_1) \times (a_2, b_2) \times \cdots \times (a_n, b_n) \times \mathbb{R} \times \cdots$ (with $a_i, b_i \in \mathbb{R} \cup \{-\infty, \infty\}$).

Also, if (S, σ) and (T, τ) are topological spaces, then $f : S \to T$ is continuous if $f^{-1}(\text{open set}) = \text{open set}$. Additionally, if (S, σ) is induced by a metric, then (T, τ) is induced by a metric, and the reason is due to the fact that continuity of functions is equivalent to the condition that

$$x_i \to x \implies f(x_i) \to f(x)$$

which is a concept in metric spaces.

Now, for $\mathbb{R} \times \mathbb{R} \times \cdots$, it turns out the product topology is induced by the metric

$$d(x, y) = \sum_{i=1}^{\infty} \frac{|x_i - y_i| \wedge 1}{2^i}$$

For some reason, this implies that coordinate wise convergence implies convergence in the product topology, but the same idea doesn't work for the box topology.

The nice thing about the product topology is that if $K_1, K_2, \cdots$ are compact, then $K_1 \times K_2 \times \cdots$ is compact in the product topology. We will use sequential compactness to show this. First, let $(x_{n,1}, x_{n,2}, \cdots)$ for each n be elements of $K_1 \times K_2 \times \cdots$. Now, since K_1 is compact, we can pass the sequence $x_{n,1}$ the a convergent subsequence $x_{n_k,1}$. Now, the trick is as follows: Since K_2, we can pass the sequence $x_{n_k,2}$ to a convergent subsequence $x_{n_{k_j},2}$. Continue this process, and denote these convergent subsequences $x_{n_k^{(1)},1}, x_{n_k^{(2)},2}, \cdots$. Now, the trick is diagonalization. Since we have that

$$n_k^{(1)} \supseteq n_k^{(2)} \supseteq n_k^{(3)} \supseteq \cdots$$

we see that the sequence formed by picking the first term in $n_k^{(1)}$, second term in $n_k^{(2)}$, etc. is eventually a subsequence of any $n_k^{(N)}$. Thus, this sequence converges in all coordinates.

9/8/26 Lecture 4: Measurability of Functions

Lemma 4.1

Let f be a mapping between two measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$. Then,

1. $\mathcal{S}' = f^{-1}(\mathcal{T})$ is a σ-field on S.
2. $\mathcal{T}' = \{B \subseteq T : f^{-1}(B) \in \mathcal{S}\}$ is a σ-field on T.

Proof: Let $A, A_1, A_2 \in \mathcal{S}'$ and let $B, B_1, B_2 \in \mathcal{T}$ so that $B_i = f^{-1}(A_i)$ and $B = f^{-1}(A)$. Then, by the fact that f distributes over complements and unions, we see that

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{S}' \text{ and } A^c \in \mathcal{S}'$$

(2) holds similarly.

Definition 4.2: Measurability of Functions

Given two measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$, a map $f : S \to T$ is said to be $\mathcal{S}\backslash\mathcal{T}$-*measurable* if $f^{-1}(\mathcal{T}) \subseteq \mathcal{S}$

For example, we see that $(\mathbb{R}, \mathcal{P}(\mathbb{R}))$ and $(\mathbb{R}, \mathcal{T})$ are measurable spaces, all functions $f : \mathbb{R} \to \mathbb{R}$ are $\mathcal{P}(\mathbb{R})\backslash\mathcal{T}$-measurable. For maps between $(\mathbb{R}, \mathcal{B})$, continuous maps are the $\mathcal{B}\backslash\mathcal{B}$-measurable functions.

Lemma 4.3

Considering a mapping f between two measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$ and let $\mathcal{C} \subseteq \mathcal{P}(T)$ with $\sigma(\mathcal{C}) = T$, then f is $\mathcal{S}\backslash\mathcal{T}$ measurable if, and only if, $f^{-1}(\mathcal{C}) \subseteq \mathcal{S}$

Proof:
($\Rightarrow$) is trivial, so we only need to show ($\Leftarrow$). Suppose that $f^{-1}(\mathcal{C}) \subseteq \mathcal{S}$.

Let $\mathcal{T}' = \{B \subseteq T : f^{-1}B \in \mathcal{S}\}$. Then, we see that $\mathcal{C} \subseteq \mathcal{T}$. Then, as $\mathcal{T}'$ is a σ-field, we see that

$$\mathcal{T} = \sigma(\mathcal{C}) \subseteq \sigma(\mathcal{T}') = \mathcal{T}'$$

Thus, as $\mathcal{T} \subseteq \mathcal{T}'$, we see that $f^{-1}(\mathcal{T}) \subseteq \mathcal{S}$.

■

Lemma 4.4

Let f be a continuous map between two topological spaces S and T with their Borel σ-fields $\mathcal{S}$ and $\mathcal{T}$. Then, f is $\mathcal{S}\backslash\mathcal{T}$ measurable.

Proof: Let $\mathcal{S}'$ and $\mathcal{T}'$ be the collection of open sets in S and T. Now, because f is continuous, we have that $f^{-1}(\mathcal{T}') \subseteq \mathcal{S}'$. Then, it is clear that $f^{-1}(\mathcal{T}') \subseteq \sigma(\mathcal{S}') = \mathcal{S}$. Then, as $\mathcal{T} = \sigma(\mathcal{T}')$, we see by Lemma 4.3 that

f is $\mathcal{S}\backslash\mathcal{T}$-measurable. ■

Notation: If $(S,\mathcal{S})$ and $(T,\mathcal{T})$ are measurable spaces and $f : S \to T$ is $\mathcal{S}\backslash\mathcal{T}$-measurable, then we just say f is measurable for brevity.

Lemma 4.5

If $(S,\mathcal{S})$, $(T,\mathcal{T})$, and $(U,\mathcal{U})$ are measurable spaces and $f : S \to T$ and $g : T \to U$ are measurable, then $h = g \circ f : S \to U$ is measurable.

Proof: Since $h^{-1} = f^{-1} \circ g^{-1}$, we see that as $g^{-1}(\mathcal{U}) \subseteq \mathcal{T}$, we see that

$$h^{-1}(\mathcal{U}) = f^{-1}(g^{-1}(\mathcal{U})) \subseteq f^{-1}(\mathcal{T}) \subseteq \mathcal{S}$$

Lemma 4.6

Consider any set of functions $f_t : \Omega \to S_t$, $t \in T$ where $(\Omega,\mathcal{A})$ and $(S_t,\mathcal{S}_t)$, $t \in T$ are measure spaces, and define $f = (f_t) : \Omega \to \bigtimes_t S_t$. Then, f is $\mathcal{A}\backslash\bigotimes_t \mathcal{S}_t$-measurable if, and only if, f_t is $\mathcal{A}\backslash\mathcal{S}_t$-measurable for all $t \in T$.

Proof: For the ($\Leftarrow$) direction, consider the family $\mathcal{C}$ of cylinder sets $A_t \times \bigtimes_{s\neq t} S_s$ for arbitrary $t \in T$ and $A_t \in \mathcal{S}_t$. Then, as

$$f^{-1}(A_t \times \bigtimes_{s\neq t} S_t) = f_t^{-1}(A_t) \cap \bigcap_{s\neq t} f_s^{-1}(S_s) \in \mathcal{A}$$

as $\mathcal{A}$ is closed under intersections. Thus, we see that $f^{-1}(\mathcal{C}) \subseteq \mathcal{A}$, which proves the theorem.

For the ($\Rightarrow$) direction, we see that if $C \in \mathcal{S}_t$, we want to show that $f_t^{-1}(C) \in \mathcal{A}$. Well, because $C \times \bigtimes_{s\neq t} S_t \in \bigotimes_t \mathcal{S}_t$, we see that

$$f^{-1}(C \times \bigtimes_{s\neq t} S_t) = f_t^{-1}(C) \cap \Omega = f_t^{-1}(C) \in \mathcal{A}$$

which was to be proved. ■

Now, if $f : \Omega \to S$ and S has a σ-field $\mathcal{S}$, then $f^{-1}(\mathcal{S})$ is a σ-field on Ω, called $\sigma(f)$. This definition makes sense as it is, in some sense, the smallest σ-field that makes f measurable.

Now, assume that f_t are mappings on Ω into some measurable space $(S_t,\mathcal{S}_t)$, and let $f = (f_t)$. In Ω, let $\sigma(f) = \sigma\{f_t : t \in T\}$ be the smallest σ-field in Ω that makes all the f_t measurable (making f measurable by Lemma 4.6). That is

$$\sigma(f) = \bigcap\{\mathcal{A} : f_t \text{ is } \mathcal{A}\backslash\mathcal{S}_t \text{ measurable for all } t\}$$

With this notation, we see that f_t is measurable with respect to the σ-field $\mathcal{A}$ in Ω if, and only if, $\sigma(f) \subseteq \mathcal{A}$. Note that, in fact, this coincides with the σ-field generated by all of the $\sigma(f_t)$'s. That is

$$\sigma(f) = \sigma\left(\bigcup_t \sigma(f_t)\right)$$

For functions on or into $\mathbb{R}^d$, we assume measurability is with respect to the Borel σ-field $\mathcal{B}^d = \mathcal{B}(\mathbb{R}^d)$. Thus, a real-valued function f on a space $(\Omega, \mathcal{A})$ is measurable iff $\{\omega : f(\omega) \leq x\} \in \mathcal{A}$ for all x (this is because these sublevel sets generate $\mathcal{B}$).

Similarly, for f on $\overline{R} = [-\infty, \infty]$ or $\overline{R}_+ = [0, \infty]$ (compactifications of $\mathbb{R}$ and $\mathbb{R}_+$) checking the above condition works to check measurability as well. Note that this is because $\mathcal{B}(\overline{\mathbb{R}}) = \sigma(\mathcal{B} \cup \{\pm\infty\})$ and $\mathcal{B}(\overline{\mathbb{R}}_+) = \sigma\{\mathcal{B}(\mathbb{R}_+), \infty\}$.

9/10/26 Lecture 5: Measurable Functions (cont.)

Definition 5.1: Indicator and Simple Functions

For any $A \subseteq \Omega$, define the *associated indicator function* $1_A : \Omega \to \mathbb{R}$ to be equal to 1 on A, and 0 on A^c. Finite linear combinations of indicator functions, such as

$$f = c_1 1_{A_1} + \cdots + c_n 1_{A_n}$$

are called *simple functions*.

Taking $c_1, \ldots, c_n$ to be the distinct nonzero values of f, we can always define $A_k = f^{-1}\{c_k\}$ and write

$$f = c_1 1_{A_1} + \cdots c_n 1_{A_n}$$

for disjoint $A_1, \ldots, A_n$. Then, we see that f is measurable with respect to a σ-field $\mathcal{A}$ if, and only if, $A_1, \ldots, A_n \in \mathcal{A}$. Notice that the representation of f above is unique if we force simple functions to be represented as finite linear combinations "disjoint" indicators.

Lemma 5.2

Let $f_1, f_2, \ldots,$ be measurable functions from $(\Omega, \mathcal{A}) \to \overline{\mathbb{R}}$. Then, $\sup_n f_n$, $\inf_n f_n$, $\limsup_n f_n$ and $\liminf_n f_n$ are all measurable.

Proof: Notice that we can write

$$(\sup_n f_n)^{-1}[-\infty, t] = \bigcap_n f_n^{-1}[-\infty, t] \in \mathcal{A}$$

so we see that $\sup_n f_n$ is measurable. Now, since $(-f_n)^{-1}[-\infty, x] = f_n^{-1}[-x, \infty]$, we see that $-f_n$ is measurable, so as $\inf_n f_n = -\sup_n(-f_n)$, we see that $\inf_n f_n$ is also measurable. Finally, as

$$\limsup_n f_n = \inf_n \sup_{k \geq n} f_k \quad \text{and} \quad \liminf_n f_n = \sup_n \inf_{n \geq k} f_k$$

we see that these are also measurable.

■

Lemma 5.3

Let $f_1, f_2, \ldots,$ be measurable functions from $(\Omega, \mathcal{A})$ into some metric space (S, ρ). Then,

1. $\{\omega : f_n(\omega) \text{ converges}\} \in \mathcal{A}$ when S is complete.
2. $f_n \to f$ on Ω implies that f is measurable.

Proof: To prove (1), we see that S is complete implies that $f_n(\omega)$ coverges if, and only if, $f_n(\omega)$ is Cauchy. Thus, we see that $f_n(\omega)$ converges iff

$$\lim_{n \to \infty} \sup_{m \geq n} \rho(f_m, f_n) = 0$$

Since $\rho(f_m, f_n)$ is measurable as $(f_m, f_n) : \Omega \to S \times S$ is measurable (both f_n and f_m are measurable), and $\rho : S \times S \to \mathbb{R}^{\geq 0}$ is measurable as $\rho^{-1}(\infty, a) = \bigcup_{s \in S}(\{s\} \times B(s, a))$, which is in $\mathcal{B}(S \times S)$. Thus, we see that

$\rho(f_n, f_m)$ is measurable, so the LHS is measurable by Lemma 5.2. Thus, we see that

$$\{\omega : f_n(\omega) \text{ converges}\} = (\lim_{n\to\infty} \sup_{m\geq n} \rho(f_m, f_n))^{-1}\{0\} \in \mathcal{A}$$

To prove (2), we see that if $f_n \to f$, then $g \circ f_n \to g \circ f$ for any continuous $g : S \to \mathbb{R}$. Thus, by Lemma 5.2 and Lemma 4.4, we see that $g \circ f$ is measurable. Now, for any open set $G \subseteq S$, we see that we can find continuous functions g_n so that $g_n \uparrow 1_G$, which implies that $1_G \circ f$ is measurable. Thus, we see that $f^{-1}(G) \in \mathcal{A}$ for all open $G \subseteq S$, which implies that f is measurable.

Now, here is one of the most important theorems, especially going into integration.

Theorem 5.4: Simple Approximation of Measurable Functions

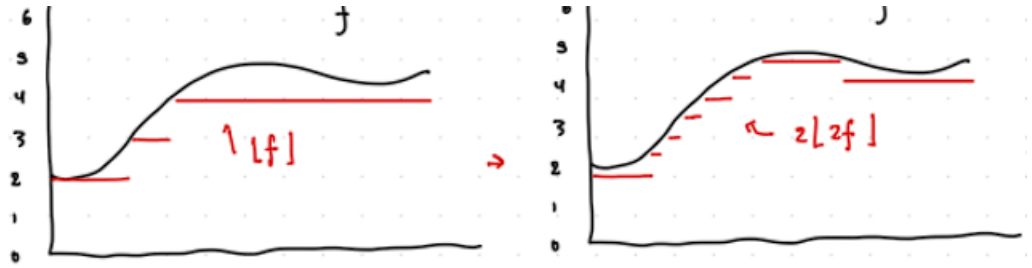
For any measurable function $f : (\Omega, \mathcal{A}) \to \mathbb{R}_+$, there exists measurable simple functions $f_1, f_2, \ldots : \Omega \to \mathbb{R}_+$ with $0 \leq f_n \uparrow f$.

Proof: Let $\lfloor g(\omega) \rfloor$ be the function $\lfloor g(\omega) \rfloor = \sum_{i\in\mathbb{N}_+} i1_{A_i}$ where $A_i = \{\omega : g(\omega) \in [i, i+1)\}$. Then, for

$$f_n(\omega) = 2^{-n}\lfloor 2^n f(\omega) \rfloor \wedge n$$

for $\omega \in \Omega$ and $n \in \mathbb{N}$ (where $\wedge$ is the minimum operator to ensure that f_n is actually a simple function by controlling it's number of terms), we can see that $f_n \uparrow f$.

We can see this construction in some sense here:



Lemma 5.5

Fix measurable functions $f, g : (\Omega, \mathcal{A}) \to \mathbb{R}$ and let $a, b \in \mathbb{R}$. Then, $af + bg$ and fg are measurable, and so is f/g if $g \neq 0$ on Ω.

Proof: If we apply Theorem 5.4 to $f_\pm = (\pm f) \vee 0$, and $g_\pm = (\pm g) \vee 0$, then we get simple functions $f_n \to f$ and $g_n \to g$.

Now, as $af_n + bg_n$ and $f_n g_n$ are simple measurable functions, and as $af_n + bg_n \to af + bg$ and $f_n g_n \to fg$, we see that $af + bg$ and fg are measurable. The same argument holds for f/g if we force $g_n \neq 0$ (by adding an "error" term that dissapears as $n \to \infty$).

Alternatively, we can prove this by writing $af + bg$, fg, and f/g as a composition $\psi \circ \phi$ where $\phi = (f, g)$

and ψ is either $\psi(x, y) = ax + by$, $\psi(x, y) = xy$ or $\psi(x, y) = x/y$. Then, it amounts to showing that ψ is measurable, which is not a difficult task.

■

9/15/26 Lecture 6: Introduction to Measures

Let $(\Omega, 2^\Omega)$ be a measurable space where Ω is the set of students in a class. Let $f : \Omega \to \{\text{male}, ttfemale\}$ and consider $\sigma(f)$.

Now, let $g : \Omega \to \{0, 1, \ldots\}$ by giving the number of dollars the student has.

Question: When is g $\sigma(f)$-measurable.

Answer: This holds if $\sigma(g) \subseteq \sigma(f)$, which holds if, and only if, there exists a map $h : \{\text{male}, \text{female}\} \to \{0, 1, 2, \ldots\}$ such that $g = h \circ f$ (this is because one direction holds, and given that $\sigma(g) \subseteq \sigma(f)$, we can construct a function h as we know each preimage of g is a preimage of f).

Definition 6.1: f-measurable

Let $f : \Omega \to X$ and $g : \Omega \to Y$ be maps. Then, we say g is *f-measurable* if g is $\sigma(f)$-measurable.

Definition 6.2: Borel Isomorphims and Borel Spaces

Two measurable spaces S and T are *Borel isomorphic* if there exists a bijection $f : S \to T$ such that f and f^{-1} are measurable. A measurable space that is Borel isomorphic to a Borel subset of $[0, 1]$ is called a *Borel space*.

Definition 6.3: Polish Spaces

A *Polish space* is a topological space that has a separable and complete metrization.

Lemma 6.4

Every Polish space with its Borel σ-field is Borel.

Note: The reason these definitions above are useful is that many spaces are Borel spaces, and it is often convenient to prove facts about $\mathbb{R}$ and then extend to other spaces via Borel isomorphisms.

Here's an example to show how common Borel spaces are. Consider $S = ([0, 1], \mathcal{B}([0, 1]))$ and $T = \{0, 1\}^{\mathbb{N}}$ with it's product σ-field. They look quite different, but if we consider $f : [0, 1] \to T$ by $f(x) = (x)_2$ (it's binary representation), then choosing representations carefully, it turns ou f is a Borel isomorphism.

Using this same argument, we see that $T \times T$ is Borel isomorphic to $[0, 1] \times [0, 1]$, and as $T \times T$ is Borel isomorphic to T by interleaving binary strings, we see that $[0, 1]$ is Borel isomorphic to $[0, 1]^2$. Thus by induction, we see that $[0, 1]$ is Borel isomorphic to $[0, 1]^n$, which is quite odd!

An example about Polish spaces is $(0, 1)$ with the usual topology. This may seem wrong at first since the normal metric isn't complete in $(0, 1)$, but there does exist some metric that works (Hint: Identify $(0, 1)$ with $\mathbb{R}$). It can also be shown that ℓ^∞ is Polish and is Borel with it's Borel σ-algebra.

Now, going back to measures, here is a generalization of the fact we saw above.

Lemma 6.5: Doob-Dynkin Lemma

Fix two measurable functions f and g on Ω into measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$ where the former, $(S, \mathcal{S})$ is Borel. Then, f is g-measurable if, and only if, there exists a measurable $h : T \to S$ such that $f = h \circ g$.

Now, after all of this rigamarole, we define what a measure is.

Definition 6.6: Measures

Given any measurable space $(\Omega, \mathcal{A})$, a map $\mu : \mathcal{A} \to \overline{\mathbb{R}}_+$ is *countably additive* if

$$\mu\left(\bigcup_{k\geq 1} A_k\right) = \sum_{k\geq 1} \mu(A_k) \text{ with } A_1, \ldots \in \mathcal{A} \text{ disjoint}$$

A *measure* on $(\Omega, \mathcal{A})$ is a map $\mu : A \to \overline{\mathbb{R}}_+$ such that $\mu(\emptyset) = 0$ and that μ is countably additive. A triple $(\Omega, \mathcal{A}, \mu)$ is a *measure space*.

Lemma 6.7

Let μ be a measure on $(\Omega, \mathcal{A})$ and assume that $A_1, A_2, \ldots \in \mathcal{A}$. Then,

1. $A_n \uparrow A$ implies $\mu(A_n) \uparrow \mu(A)$.
2. $A_n \downarrow A$ with $\mu(A_1) < \infty$ implies $\mu(A_n) \downarrow \mu(A)$

Proof: Apply countable additivity to $D_n = A_n \backslash A_{n-1}$ (defining $A_0 = \emptyset$), and for (ii) apply (i) to $B_n = A_1 \backslash A_n$, using the fact that $A_1 \backslash A_n \uparrow A_1 \backslash A$, so $-\mu(A_n) \uparrow -\mu(A)$ (this step requries A_1 to have finite measure). ■

Proposition 6.8

For any measures $\mu_1, \mu_2, \ldots$ on $(\Omega, \mathcal{A})$ and constants $c_1, c_2, \ldots, \geq 0$, the sum $\mu = \sum_n c_n \mu_n$ is again a measure.

Proof: We first want to show that $\sum_i \sum_j c_{ij} = \sum_j \sum_i c_{ij}$ for $c_{ij} \geq 0$, $i, j \in \mathbb{N}$. Since

$$\sum_i \sum_j c_{ij} \geq \sum_{i\leq m} \sum_{j\leq n} c_{ij} = \sum_{j\leq n} \sum_{j\leq m} c_{ij}$$

Taking $n, m \to \infty$ (one at a time), we see that $\sum_i \sum_j c_{ij} \geq \sum_j \sum_i c_{ij}$. By symmetry, equality holds.

Now, if $A_1, A_2, \ldots \in \mathcal{A}$ are disjoint, then we see by countable additivity of the components,

$$\begin{aligned}
\mu\left(\bigcup_{k\geq 1} A_k\right) &= \sum_n c_n \mu_n \left(\bigcup_{k\geq 1} A_k\right) \\
&= \sum_n c_n \sum_{k\geq 1} \mu_n(A_k) \\
&= \sum_{k\geq 1} \sum_n c_n \mu_n(A_k) \\
&= \sum_{k\geq 1} \mu(A_k)
\end{aligned}$$

■

9/17/26 Lecture 7: The Lebesgue Integral

Corollary 7.1

Let $\mu_1, \mu_2, \dots$ be a measure on $(\Omega, \mathcal{A})$ such that $\mu_n \uparrow \mu$ or $\mu_n \downarrow \mu$ with μ_1 bounded. Then, μ is a measure on $(\Omega, \mathcal{A})$.

Proof: For increasing, use the fact that if $0 \leq c_{nj} \uparrow c_j$ for all j, then $\sum_j c_{nj} \uparrow \sum_j c_j$, which allows us to conclude μ satisfies measure properties.

For the decreasing direction, show that $\mu_1 - \mu_n$ is a measure and that $\mu_1 - \mu_n \uparrow \mu_1 - \mu$.

■

Lemma 7.2

Let μ and ν be bounded measures on some measurable space $(\Omega, \mathcal{A})$ and let $\mathcal{C}$ be a π-system in Ω such that $\Omega \in \mathcal{C}$ and $\sigma(\mathcal{C}) = \mathcal{A}$. Then $\mu = \nu$ if, and only if, $\nu(A) = \mu(A)$ for all $A \in \mathcal{C}$.

Proof: Let $\mathcal{D}$ be the collection of all $A \in \mathcal{A}$ such that $\mu(A) = \nu(A)$. Then, show that $\mathcal{D}$ is a λ-system, so as $\mathcal{C}$ is a π-system contained in $\mathcal{D}$ we see that $\sigma(\mathcal{C}) \subseteq \mathcal{D}$ by the $\pi - \lambda$ theorem. Thus, we see that $\mathcal{A} \subseteq \mathcal{D}$, so $\mu = \nu$.

■

Definition 7.3: Push-Forward Measures

Let $f : S \to T$ be measurable on two spaces $(S, \mathcal{S}), (T, \mathcal{T})$. Given any measure μ on $(S, \mathcal{S})$, we can define a measure $\mu \circ f^{-1}$ on $(T, \mathcal{T})$ by

$$(\mu \circ f^{-1})(B) = \mu(f^{-1}(B)) = \mu\{s \in S \,:\, f(s) \in B\}$$

for $B \in \mathcal{T}$. We call $\mu \circ f^{-1}$ the *push-forward measure of μ by f*.

An example that motivates this definition is if we let $(\Omega, 2^\Omega)$ be the class of students, and let μ be a measure defined by $\mu(\{s\}) = \frac{1}{|\Omega|}$. That is, μ is uniform on the collection of students and gives a proportion. Then, if $f : \Omega \to \{\text{male, female}\}$, then we see that the push-forward measure has the property that

$$\mu \circ f^{-1}(\{\text{male}\}) = \frac{\#\text{ of males}}{\#\text{ of students}}$$

so it gives the measure you would expect!

Now, we want to define the integral of a real valued measurable function f on some measure space $(\Omega, \mathcal{A}, \mu)$ which we denote in any of these ways

$$\mu f = \int f d\mu = \int f(\omega) d\mu(\omega)$$

We will build up. First, we assume that f is simple and nonnegative, so hence of the form $f = c_1 1_{A_1} + \cdots + c_n 1_{A_n}$ for some $z \in \mathbb{Z}_+$, $A_1, \ldots, A_n \in \mathcal{A}$ and $c_1, \ldots, c_n \in \mathbb{R}_+$. Then, we define

$$\mu f = c_1 \mu(A_1) + \cdots + c_n \mu(A_n)$$

using the convention that $0 \cdot \infty = 0$. It can be shown that μf is independent of choice of representation of f. It can also be shown that, on such simple functions, $f \mapsto \mu f$ is linear and non-decreasing; that is

$$\mu(af + bg) = a\mu f + b\mu g \quad (a, b \geq 0)$$
$$f \leq g \implies \mu f \leq \mu g$$

Now, let's extend this to more general functions. For any non-negative measurable function, let $\mu f = \lim_n \mu f_n$ where $f_1, f_2, \ldots$ are measurable simple functions with $0 \leq f_n \uparrow f$ (recall these exist!). Now, the problem is that this definition may not be unique; does it agree with any other increasing $0 \leq g_n \uparrow f$?

Lemma 7.4

Let $f \geq 0$ be measurable on a measure space $(\Omega, \mathcal{A}, \mu)$ and let $f_1, f_2, \ldots$ and g be simple measurable functions such that $f_n \uparrow f$ and $0 \leq g \leq f$. Then $\lim_n \mu f_n \geq \mu g$.

Proof: By linearity of μ, we just need to consider when $g = 1_A$. Fix $\epsilon > 0$ and let

$$A_n = \{\omega \in A : f_n(\omega) \geq 1 - \epsilon\}$$

Now, as $A_n \uparrow A$ (since $f \geq 1_A$), then, as

$$\mu f_n \geq (1 - \epsilon)\mu(A_n) \text{ and } \mu(A_n) \uparrow \mu(A)$$

We see that as $\lim_n \mu f_n \geq (1 - \epsilon)\mu(A) = (1 - \epsilon)\mu g$. Thus, taking $\epsilon \to 0$, we see that $\lim_n \mu f_n \geq \mu g$.

■

Corollary 7.5

If $f_1, f_2, \ldots, g_1, g_2, \ldots \geq 0$ are measurable simple functions with $f_n \uparrow f$ and $g_n \uparrow f$, then $\lim_n \mu f_n = \lim_n \mu g_n$.

Proof: Since $g_n \leq f$ for all n, we see that $\lim_n \mu f_n \geq \mu g_n$ for all n, so $\lim_n \mu f_n \geq \lim_n \mu g_n$. Then, by symmetry equality holds.

■

Now, we know how to extend μ to all measurable functions $f : \Omega \to \overline{\mathbb{R}}_+$ with monotonicity and linearity properties. Now, we will prove some useful lemmas about this integral.

Theorem 7.6: Monotone Convergence Theorem

Let $f, f_1, f_2, \ldots$ be measurable functions on $(\Omega, \mathcal{A}, \mu)$ with $0 \leq f_n \uparrow f$. Then, $\mu f_n \uparrow \mu f$.

Proof: For each n, choose simple measurable $0 \le g_{n,k} \uparrow f_n$. Now, let

$$h_{n,k} = g_{1,k} \vee \cdots \vee g_{n,k}$$

We see that $h_{n,k}$ has the same properties as $g_{n,k}$ since it's positive and also non decreasing in both indices as

$$h_{n,k} \le h_{n+1,k} \text{ since } g_{1,k} \vee \cdots \vee g_{n,k} \le g_{1,k} \vee \cdots \vee g_{n,k} \vee g_{n+1,k}$$

and

$$h_{n,k} \le h_{n,k+1} \text{ since } g_{1,k} \vee \cdots \vee g_{n,k} \le g_{1,k+1} \vee \cdots \vee g_{n,k+1}$$

Also, we see that $h_{n,k} \uparrow f_n$ as $k \to \infty$ $g_{i,k} \le f_i \le f_n$ for $i < n$. Since $h_{k,k} \le f$, we see that

$$f \ge \lim_{k\to\infty} h_{k,k} \ge \lim_{k\to\infty} h_{n,k} = f_n$$

However, as $f_n \uparrow f$, we see that $0 \le h_{k,k} \uparrow f$. Then, as $\mu f_n \le \mu f$ for all n,

$$\mu f = \lim_{k\to\infty} \mu h_{k,k} \le \lim_{k\to\infty} \mu f_k \le \mu f$$

So we see that $\mu f_k \uparrow \mu f$.

■

Lemma 7.7: Fatou's Lemma

For any measurable function $f_1, f_2, \cdots \ge 0$ on $(\Omega, \mathcal{A}, \mu)$, we have

$$\liminf_{n\to\infty} \mu f_n \ge \mu \liminf_{n\to\infty} f_n$$

Proof: Since $f_m \ge \inf_{k\ge n} f_k$ for all $m \ge n$ so we have

$$\inf_{k\ge n} \mu f_k \ge \mu \inf_{k\ge n} f_n$$

since $\mu f_m \ge \mu \inf_{k\ge n} f_n$ for all $m \ge n$. Letting $n \to \infty$,

$$\liminf_{k\to\infty} \mu f_k \ge \lim_{n\to\infty} \mu \inf_{k\ge n} f_k = \mu \liminf_{k\to\infty} f_k$$

by the Monotone Convergence Theorem, and as $\inf_{k\ge n} f_k \uparrow \liminf_{k\to\infty} f_k$.

■

Definition 7.8: Integrability

A measurable function f on $(\Omega, \mathcal{A}, \mu)$ is said to be *integrable* if $\mu|f| < \infty$. Then, we can write f as the difference of two non-negative integrable functions g and h $f = g - h$, and define $\mu f = \mu g - \mu h$.

Typically, we choose $f = f_+ - f_-$ with $f_\pm = (\pm f) \vee 0$, but it can be shown that this is independent of choice of representation, and μf is still linear (now allowing all coefficients!) and monotone.

9/24/26 Lecture 8: Properties of Integrals

Theorem 8.1: Dominated Convergence Theorem

Let $f, f_1, f_2, \ldots$ and $g, g_1, g_2, \ldots$ be measurable functions on $(\Omega, \mathcal{A}, \mu)$ with $|f_n| \leq g_n$ for all n and such that $f_n \to f$, $g_n \to g$ and $\mu g_n \to \mu g < \infty$. Then, $\mu f_n \to \mu f$.

Proof: Consider the collection of functions $g_n \pm f_n \geq 0$. We then see that

$$\mu g + \liminf_{n\to\infty} (\pm \mu f_n) = \liminf_{n\to\infty} \mu(g_n \pm f_n) \geq \mu(g \pm f) = \mu g \pm \mu f$$

by Fatou's lemma. Subtracting $\mu g < \infty$ from both sides, we see that

$$\liminf_{n\to\infty} (\pm \mu f_n) \geq \pm \mu f$$

Thus, we see that

$$\mu f \leq \liminf_{n\to\infty} \mu f_n$$

and

$$-\mu f \leq \liminf_{n\to\infty} -\mu f_n = -\limsup_{n\to\infty} \mu f_n \implies \mu f \geq \limsup_{n\to\infty} \mu f_n$$

Thus, we have that

$$\mu f \leq \liminf_{n\to\infty} \mu f_n \leq \limsup_{n\to\infty} \mu f_n \leq \mu f$$

Thus, we must have that $\mu f = \liminf_{n\to\infty} \mu f_n = \limsup_{n\to\infty} \mu f_n$, so

$$\mu f = \lim_{n\to\infty} \mu f_n$$

■

Lemma 8.2: Substitution Lemma

Consider a measure space $(\Omega, \mathcal{A}, \mu)$, a measurable space $(S, \mathcal{S})$ and two measurable mappings $f : \Omega \to S$ and $g : S \to \mathbb{R}$. Then,

$$\mu(g \circ f) = (\mu \circ f^{-1})(g)$$

whenever either side exists; that is, when one side exists then so does the other and they are equal.

Thus, the integral of $g \circ f$ with respect to μ is equal to the integral of g with respect to the push-forward measure $\mu \circ f^{-1}$.

Proof: If g is an indicator, then the above equality reduces to the definition of $\mu \circ f^{-1}$, as $(\mu \circ f^{-1})(1_A) = (\mu \circ f^{-1})(A) = \mu(f^{-1}(A))$, and similarly, $\mu(1_A \circ f) = \mu(f^{-1}(A))$, as $1_A \circ f = 1_{f^{-1}(A)}$.

From here, we use linearity and monotone convergence to extend to any measurable function $g \geq 0$. For a general g, it follows that $\mu\,|g \circ f| = (\mu \circ f^{-1})\,|g|$, and so the integrals on both sides of the equality exist at the same time. When they do, we can get equality using the definition of an integral of a general function.

Example

Let $\Omega = \{\text{meteroite strikes}\}$ and let S^2 be the 2-dimensional sphere. Let $f(\omega) = \text{location of meteorite strike}$ and $g(s)$ be the distance of s from the equator of S. Then $\mu(g \circ f)$ is the probability a meteor strikes a given distance from the equator, and $(\mu \circ f^{-1})(g)$ also tells you the probability a meteor strikes a given distance from the equator.

Definition 8.3: Null Sets

Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. We say that $A \in \mathcal{A}$ is said to be a *μ-null set* or a *null set* if $\mu(A) = 0$. A relationship between functions on Ω is said to hold *almost everywhere* with respect to μ, denoted *a.e.* μ or *μ-a.e.*, if it holds for all $\omega \in \Omega$ outside some μ-null set.

An example of measure 0 Borel set is the Cantor set K, which consists of elements of $[0, 1]$ that only have 0's and $2'$ in their ternary expansion. Notice that $(K, \mathcal{B}(K))$ is Borel-isomorphic to $([0, 1], \mathcal{B}([0, 1]))$, by mapping binary sequences to these such ternary sequences.

Then, as $\mathcal{B}([0, 1]) \neq 2^{[0,1]}$, by Vitali, we must have that there are some subsets of $\mathcal{B}(K)$ that are not Borel. Thus, there do exists subsets of sets of measure zero that are not measurable! However, we can still assign a measure for such sets that extends the Lebesgue measure to these sets; namely 0.

Lemma 8.4: Null Sets and Functions

For any measurable function $f \geq 0$ on some measure space $(\Omega, \mathcal{A}, \mu)$, then $\mu f = 0$ iff $f = 0$ μ-a.e.

Proof: This is obvious when f is simple. In the general case, we may choose some simple measurable functions f_n with $0 \leq f_n \uparrow f$, and note that $f = 0$ a.e. if, and only if, $f_n = 0$ a.e. for every n if, and only if, $\mu f_n = 0$ for all n. Thus, as the integrals converge to μf, we see that this condition is equivalent to saying $\mu f = 0$.

Definition 8.5: μ-completions

Given a measure space $(\Omega, \mathcal{A}, \mu)$ and a σ-field $\mathcal{F} \subseteq \mathcal{A}$, define the *$\mu$-completion* of $\mathcal{F}$ in $\mathcal{A}$ as the σ-field $\mathcal{F}^\mu = \sigma(\mathcal{F}, \mathcal{N}_\mu)$, where $\mathcal{N}_\mu$ denotes the class of all subsets of arbitrary μ-null sets in $\mathcal{A}$.

9/29/26 Lecture 9: Lebesgue-Fubini-Tonelli-Hölder-Minkowski

Lemma 9.1: Completion

Consider a measure space $(\Omega, \mathcal{A}, \mu)$, a σ-field $\mathcal{F} \subseteq \mathcal{A}$, and a Borel space $(S, \mathcal{S})$. Then, a function $f : \Omega \to S$ is $\mathcal{F}^\mu$-measurable if, and only if, there exists some $\mathcal{F}$-measurable function g such that $f = g$ a.e. μ.

Proof: Let's start with indicators. If $\mathcal{G}$ is the set of $A \subseteq \Omega$ such that $A \Delta B \in \mathcal{N}_\mu$ for some $B \in \mathcal{F}$. Then, we have that $A \backslash B$ and $B \backslash A$ are in $\mathcal{N}_\mu$ (they are subset of a null set), which implies that $\mathcal{G} \subseteq \mathcal{F}^\mu$. This is because we know that $B \cup A \backslash B = A \cup B \in \mathcal{F}^\mu$ and $A \cup B \backslash (B \backslash A) = A \in \mathcal{F}^\mu$. Conversely, $\mathcal{F}^\mu \subseteq \mathcal{G}$ since both $\mathcal{F}$ and $\mathcal{N}_\mu$ are trivially ($\mathcal{F}$ is trivial, for $\mathcal{N}_\mu$, consider $A \Delta \emptyset$) contained in $\mathcal{G}$, and it can be shown that $\mathcal{G}$ is a σ-field.

Then, we see that $A \in \mathcal{F}^\mu$ if, and only if $1_A = 1_B$ almost everywhere for some $B \in \mathcal{F}$. When, we may choose some simple $\mathcal{F}^\mu$-measurable functions f_n such that $0 \leq f_n \uparrow f$. By our result above there exists $\mathcal{F}$-measurable functions g_n such that $f_n = g_n$ almost everywhere. Since a countable union of null sets is a null set, we see that $g = \limsup_n g_n$ has the property that we want.

■

Definition 9.2: σ-finiteness

A measure μ on a measurable space $(\Omega, \mathcal{A})$ is *σ-finite* if $\Omega = \bigcup_{i=1}^\infty A_n$ with $A_i \cap A_j = \emptyset$ for $i \neq j$ such that $\mu A_n < \infty$ for all n. Equivalently, μ is σ-finite if, and only if, there is a measurable strictly positive function f such that $\mu f < \infty$.

The following results answer when

$$\int_S \left(\int_T f(s,t) \nu(t) \right) \mu(s) = \int_T \left(\int_S f(s,t) \mu(s) \right) \mu(t)$$

First, we start with a lemma to show that the statement above even makes sense at all.

Lemma 9.3: Sections

Fix two measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$, a measurable function $f : S \times T \to \mathbb{R}_+$, and a σ-finite measure μ on S. Then,

1. $f(s,t)$ is $\mathcal{S}$ measurable in $s \in S$ for each $t \in T$.
2. $\int f(s,t) \mu(ds)$ is $\mathcal{T}$-measurable in $t \in T$.

Proof: We may assume that μ is bounded. Both statements are obvious when $f = 1_A$ with $A = B \times C$ for some $B \in \mathcal{S}$ and $C \in \mathcal{T}$. This extends by a monotone class argument to any indicator functions of sets in $\mathcal{S} \otimes \mathcal{T}$. The general case then follows by linearity and monotone convergence.

■

Theorem 9.4: Lebesgue-Fubini-Tonelli Theorem

For any σ-finite measure spaces $(S, \mathcal{S}, \mu)$ and $(T, \mathcal{T}, \nu)$, there exists a unique measure $\mu \otimes \nu$ on $(S \times T, \mathcal{S} \otimes \mathcal{T})$ satisfying

$$(\mu \otimes \nu)(B \times C) = \mu(B) \cdot \nu(C), \quad B \in \mathcal{S}, C \in \mathcal{T}$$

Furthermore, for any non-negative measureable function $f : S \times T \to \mathbb{R}_+$,

$$(\mu \otimes \nu)v = \int \mu(ds) \int f(s,t)\nu(dt) = \int \nu(dt) \int f(s,t)\mu(ds)$$

The last relation remains valid for any measurable function $f : S \times T \to \mathbb{R}$ with $(\mu \otimes \nu)\,|f| < \infty$.

Note that these integrals are well defined by the lemma above, and although the inner integrals $\nu f(s, \cdot)$ and $\mu f(\cdot, t)$ may fail to exist on some null sets in S and T, when everything works, everything agrees.

Proof: By the lemma above, we can define

$$(\mu \otimes \nu)(A) = \int \mu(ds) \int 1_A(s,t)\nu(dt), \quad A \in \mathcal{S} \otimes \mathcal{T}$$

which is a clearly a measure on $S \times T$. By a monotone class argument, there can be at most one measure (Caratheodory Extension Theorem).

In particular, if we interchange the integrals, from the same monotone class argument, we see that the corresponding measure $\nu \otimes \mu$ also agrees with $\mu \otimes \nu$. Thus, we see that the statement in the theorem about integrals holds for indicator functions. Then, by linearity and monotone convergence, it holds for arbitrary measurable functions $f \geq 0$.

Now, for the general case, we see that for any measurable f, statement about integrals holds for $|f|$. Then, if $(\mu \otimes \nu)\,|f| < \infty$, it follows that $N_S = \{s \in S : \nu\,|f(s, \cdot)| = \infty\}$ is a μ-null set in S (as otherwise we would get infinity for our integral), and similarly $N_T = \{t \in T : \mu\,|f(\cdot, t)| = \infty\}$ is also a ν-null set. Then, we can define $f(s,t)$ to be zero when $s \in N_S$ or $t \in N_t$. This new function has the same integral as the previous function.

Then, the statement about integrals follows for f by subtraction of the formulas for f_+ and f_-.

Now, we move on to L^p spaces!

Definition 9.5: L^p Spaces

Given a measure space $(\Omega, \mathcal{A}, \mu)$ and a $p > 0$, write $L^p = L^p(\Omega, \mathcal{A}, \mu)$ for the class of all measurable functions $f : \Omega \to \mathbb{R}$ with

$$\|f\|_p := \left[\int |f|^p \, d\mu\right]^{1/p} < \infty$$

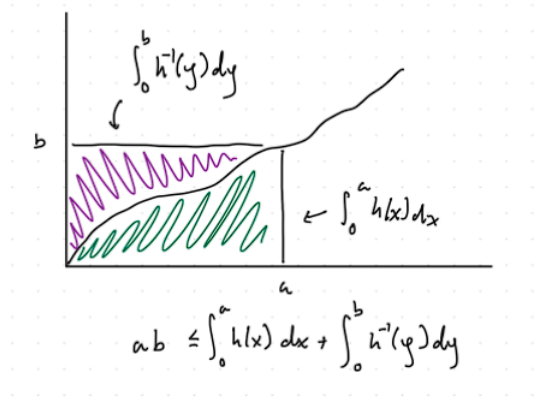
Now, here are some very useful inequalities:

Theorem 9.6: Hölder and Minkowski Inequalities

For any measurable functions f and g on some measure space $(\Omega, \mathcal{A}, \mu)$, we have

1. $\|fg\|_r \leq \|f\|_p \|g\|_q$ for all $p, q, r > 0$ with $p^{-1} + q^{-1} = r^{-1}$.
2. $\|f+g\|_p \leq \|f\|_p + \|g\|_p$ for all $p \geq 1$, and $\|f+g\|_p^p \leq \|f\|_p^p + \|g\|_p^p$ for $0 < p < 1$.

Proof: For (1), it is clearly enough to take $r = 1$ (replace f and g with f^r and g^r) and $\|f\|_p = \|g\|_q = 1$ (replace f and g with their normalized versions, and if norm 0, inequality is obvious). The relation $p^{-1} + q^{-1} = 1$ implies $(p-1)(q-1) = 1$, so the equations $y = x^{p-1}$ and $x = y^{q-1}$ are equivalent for $x, y \geq 0$. Thus, $(\cdot)^{p-1}$ and $(\cdot)^{q-1}$ are inverses. By Young's inequality, which is depicted below:



we have that,

$$|fg| \leq \int_0^{|f|} x^{p-1} dx + \int_0^{|g|} y^{q-1} dy = p^{-1} |f|^p + q^{-1} |g|^q$$

Thus,

$$\|fg\|_1 \leq p^{-1} \int |f|^p \, d\mu + q^{-1} \int |g|^q \, d\mu = p^{-1} + q^{-1} = 1$$

We will prove the second inequality next.

10/1/26 Lecture 10: More on L^p Spaces

Proof of Minkowski's Inequality: If $p \leq 1$, then by the concavity of x^p on $\mathbb{R}_+$, we we have

$$\|f+g\|_p^p = \mu(|f+g|^p) \leq \mu((|f|+|g|)^p) \leq \mu(|f|^p) + \mu(|g|^p) = \|f\|_p^p + \|g\|_p^p$$

Remark: This is because if h is concave, non-decreasing, and $h(0) = 0$, then $h(a+b) \leq h(a) + h(b)$, and this true about x^p for $0 < p \leq 1$. Now, for $p > 1$, then we see that using Hölder's inequality with $r = 1$ and $q = \frac{p}{p-1}$,

$$\begin{aligned} \|f+g\|_p^p &\leq \int |f|\,|f+g|^{p-1}\,d\mu + \int |g|\,|f+g|^{p-1}\,d\mu && \text{(factoring)} \\ &\leq \|f\|_p \|f+g\|_p^{p-1} + \|g\|_p \|f+g\|_p^{p-1} && \text{(Hölder's inequality)} \end{aligned}$$

Thus, dividing both sides by $\|f+g\|_p^{p-1}$ (if this is zero, inequality is obvious), we get

$$\|f+g\|_p \leq \|f\|_p + \|g\|_p$$

■

Because of this above results, this shows that $(f,g) \mapsto \|f-g\|_p^{p \wedge 1}$ is a metric on L^p, as it satisfies all the requirements. Thus, we can say that $g_n \to g$ in L^p if $\|g_n - g\|_p^{p-1} \to 0$. This also let's us define Cauchy sequences in the normal way, which is used in the next lemma.

Lemma 10.1: Completeness

Let (f_n) be a Cauchy sequence in L^p, where $p > 0$. Then $\|f_n - f\|_p \to 0$ for some $f \in L^p$.

Proof: Choose a subsequence $(n_k) \subseteq \mathbb{N}$ with $\sum_k \|f_{n_{k+1}} - f_{n_k}\|_p^{p \wedge 1} < \infty$. By Minkowski's inequality and monotone convergence applied to finite sums, we see that

$$\|\sum_k |f_{n_{k+1}} - f_{n_k}|\|_p^{p \wedge 1} < \infty$$

so $\sum_k |f_{n_{k+1}} - f_{n_k}| < \infty$ almost everywhere. Hence, we must have that (f_{n_k}) is Cauchy almost everywhere in $\mathbb{R}$, so by the completeness of $\mathbb{R}$, we see that $f_{n_k} \to f$ for some measurable function f. Then, by Fatou's lemma, and using the fact that $f = \liminf_{k\to\infty} f_{n_k}$,

$$\begin{aligned} \|f - f_n\|_p &= \left(\int \left| \liminf_{k\to\infty} f_{n_k} - f_n \right|^p \right)^{1/p} \\ &= \left(\int \liminf_{k\to\infty} |f_{n_k} - f_n|^p \right)^{1/p} \\ &\leq \left(\liminf_{k\to\infty} \int |f_{n_k} - f_n|^p \right)^{1/p} \\ &\leq \liminf_{k\to\infty} \|f_{n_k} - f_n\|_p \\ &\leq \sup_{m \geq n} \|f_m - f_n\| \end{aligned}$$

which is assumed to go to 0 as $n \to \infty$. Thus, we see that $f_n \to f$ in L^p.

■

Lemma 10.2: Useful Inequality

If $a, b \geq 0$, then

$$(a+b)^p \leq \begin{cases} a^p + b^p & 0 < p \leq 1 \\ 2^{p-1}(a^p + b^p) & p \geq 1 \end{cases}$$

Proof: For $0 < p \leq 1$, this is obvious by the concavity of x^p for $0 < p \leq 1$. For $p \geq 1$, we know that $x \mapsto x^p$ is convex. This can be checked because the second derivative of x^p is greater than or equal to zero for all x.

Then, we see that

$$\left(\frac{a+b}{2}\right)^p = \left(\frac{a}{2} + \frac{b}{2}\right)^p \leq \frac{1}{2}a^p + \frac{1}{2}b^p$$

Then, rearranging, we get that

$$(a+b)^p \leq 2^{p-1}\left(a^p + b^p\right)$$

■

Lemma 10.3: L^p-convergence

For any $p > 0$, let $f, f_1, f_2, \ldots \in L^p$ with $f_n \to f$ almost everywhere. Then, $f_n \to f$ in L^p iff $\|f_n\|_p \to \|f\|_p$.

Proof: If $f_n \to f$ in L^p, then by Minkowski's

$$\left|\|f_n\|_p^{p \wedge 1} - \|f\|_p^{p \wedge 1}\right| \leq \|f_n - f\|_p^{p \wedge 1} \to 0$$

and so $\|f_n\|_p \to \|f\|_p$. Now, assume the other condition, define

$$g_n = 2^p\left(|f_n|^p + |f|^p\right) \quad \text{and} \quad g = 2^{p+1}|f|^p$$

Then, we see that $g_n \to g$ almost everywhere, and $\mu g_n \to \mu g < \infty$ by our hypotheses (since $g_n < (2f)^p$). Since we see that $|g_n| \geq |f_n - f|^p$, which must go to zero almost everywhere. Thus, $\|f_n - f\|_p^p = \mu|f_n - f|^p \to 0$ by the dominated convergence theorem.

■

Now, we will consider the special case where $p = q = 2$ and $r = 1$. Using Hölder's inequality, we get the special case which is known as the *Cauchy-Buniakowsky* or *Schwarz* inequality

$$\|fg\|_1 \leq \|f\|_2 \|g\|_2$$

In particular, note that for any $f, g \in L^2$, the inner product $\langle f, g\rangle = \mu(fg)$ exists (not infinity) and satisfies $|\langle f, g\rangle| \leq \|f\|_2\|g\|_2$. Quickly, to make our foundations solid, lets make sure that $\langle f, g\rangle$ is actually an inner product.

First, it is clear that $\langle f, f\rangle \geq 0$, and $\langle f, f\rangle = 0 \iff f = 0$ (a.e.). Also, $\langle fg\rangle = \langle gf\rangle$ and from linearity of the integral,

$$\langle af + bg, h\rangle = a\langle f, h\rangle + b\langle g, h\rangle$$

Bilinearity follows from symmetry. We also see that $\|f\| = \sqrt{\langle f, f\rangle} = \sqrt{\mu f^2} = \|f\|_2$, so we will interchange between using $\|\cdot\|$ and $\|\cdot\|_2$.

Going back, we see that from the bilinearity of inner products, we ge the parallelogram identity

$$\|f+g\|^2 + \|f-g\|^2 = 2\|f\|^2 + 2\|g\|^2, \quad f, g \in L^2$$

Now, we say that two functions $f, g \in L^2$ are said to be *orthogonal*, written $f \perp g$, if $\langle f, g\rangle = 0$. Orthogonality between two subsets $A, B \subseteq L^2$ implies that $f \perp g$ for all $f \in A$ and $g \in B$.

Finally, a subspace $M \subseteq L^2$ is said to be *linear* of $af + bg \in M$ for any $f, g \in M$ and $a, b \in \mathbb{R}$. Also, a subspace $M \subseteq L^2$ is said to be *closed* if $f \in M$ whenever f is the L^2-limit of a sequence in M.

Theorem 10.4: Existence of Orthononal Decomposition

Let M be a closed linear subspace of L^2. Then, any function $f \in L^2$ has any almost everywhere unique decomposition $f = g + h$ with $g \in M$ and $h \perp M$.

Proof: Let $f \in L^2$ and define $d = \inf\{\|f - g\| : g \in M\}$. Now, choose $g_1, g_2, \ldots \in M$ so that $\|f - g_n\| \to d$. Using the linearity of M, the definition fo d, we see that for $m, n \to \infty$, we first have that, as $(g_n + g_m)/2 \in M$, and

$$\|2f - g_n - g_m\| = 2\|f - \frac{g_n + g_m}{2}\|$$

which by the definition of d, implies that

$$\|2f - g_n - g_m\| \geq 2d$$

Then, the parallelogram identity using $u = f - g_m$ and $v = f - g_n$,

$$\|2f - g_m - g_n\|^2 + \|g_m - g_n\|^2 = 2\|f - g_m\|^2 + 2\|f - g_n\|^2 \leq 4d^2$$

Thus,

$$\begin{aligned} 4d^2 + \|g_m - g_n\|^2 &= (2d)^2 + \|g_m - g_n\|^2 \\ &\leq \|2f - g_m - g_n\|^2 + \|g_m - g_n\|^2 \\ &= 2\|f - g_m\|^2 + 2\|f - g_n\|^2 \to 4d^2 \end{aligned}$$

Thus, $\|g_m - g_n\| \to 0$, so the sequence (g_n) is Cauchy in L^2, which means it converges to some $g \in L^2$ and as M is closed, $g \in M$. We will finish the proof later.

■

10/6/26 Lecture 11: Useful Theorems about Integration

We will finish the proof of Theorem 10.4.

Proof of Theorem 10.4 (cont.): Now, as we see that $h := f - g$ has norm d, we see that for any $l \in M$, since $\|h + tl\| = \|f - (g - tl)\|$ and as $g - tl \in M$,

$$d^2 \leq \|h + tl\|^2 = d^2 + 2t \langle h, l \rangle + t^2 \|l\|^2, \ \forall t \in \mathbb{R}$$

Now, we have two cases. If $\langle h, l \rangle > 0$, then for $t < 0$ sufficiently close to 0 so that $|2t \langle h, l \rangle| > \left|t^2 \|l\|^2\right|$, which implies that

$$d^2 + 2t \langle h, l \rangle + t^2 \|l\|^2 < d^2$$

as $2t \langle h, l \rangle < 0$ which is a contradiction. Similarly, if $\langle h, l \rangle < 0$, then do the same thing by taking $t > 0$ sufficiently close to 0. Thus, we must have that $\langle h, l \rangle = 0$, so $h \perp M$ as desired.

To prove uniqueness, let $g' + h'$ be another decomposition with the stated properties. Then $g - g' \in M$, and also $g - g' = h' - h \perp M$, so $g - g' \perp g - g'$, which implies $\|g - g'\|^2 = \langle g - g', g - g' \rangle = 0$, so $g = g'$ almost everywhere.

■

Example

Suppose μ is a probability measure and M is the closed subspace of constant functions. This is clearly in L^2 since μ is a probability measure. Now, for $f \in L^2(\mu)$, what is the orthogonal projection of f onto M?

Thus, will be the unique constant function c such that $\|f - c\|_2^2$ is minimized. Notice that

$$\begin{aligned} \|f - c\|^2 &= \|f\|_2^2 - 2 \langle f, c \rangle + \|c\|_2^2 \\ &= \|f\|_2^2 - 2c\mu f + c^2 \end{aligned}$$

Differentiating with respect to c and setting this to zero, we see clearly that

$$c = \mu f$$

Now, what if we let $M = \{\text{functions in } L^2 \text{ that are measureable wrt to some } \sigma - \text{field } \mathcal{F} \subseteq \mathcal{A}\}$.

We see that random variables are just functions and expectations are just integrals. Thus, we will call the orthogonal projection the "conditional expectation"

Lemma 11.1: Regularity

Let μ be a bounded measure on some metric space S with Borel σ-field $\mathcal{S}$. Then,

$$\mu B = \sup_{F \subseteq B} \mu F = \inf_{G \supseteq B} \mu G, \ B \in \mathcal{S}$$

with F and G restricted to the classes of closed and open subsets of S, respectively.

Proof: For any open set G, let

$$F_n = \{x \in G : \overline{B}(x, 1/n) \subseteq G\}$$

these are closed sets $F_n \uparrow G$, so by a previous lemma, $\mu F_n \uparrow \mu G$. This proves the statement for B belonging to the π-system $\mathcal{G}$ of open sets. Letting $\mathcal{D}$ denote the class of all sets B with the stated property, we can show that $\mathcal{D}$ is a λ-system containing $\mathcal{G}$, which implies that $\mathcal{S} = \sigma(\mathcal{G}) \subseteq \mathcal{D}$.

A similar result shows that $B = \inf_{G \supseteq B} \mu G$, by showing it holds for closed sets (using open balls at each point), and then applying π-λ.

Lemma 11.2: Approximation

Given a metric space S with Borel σ-field $\mathcal{S}$ and a bounded measure μ on $(S, \mathcal{S})$ and a constant $p > 0$, the set of bounded continuous functions on S is dense in $L^p(S, \mathcal{S}, \mu)$. Thus, for any $f \in L^p$, there exist some bounded continuous functions $f_1, f_2, \ldots : S \to \mathbb{R}$ with $\|f_n - f\|_p \to 0$.

Proof: If $f = 1_A$ with $A \subseteq S$ open, then let

$$f_n(x) = (n \inf\{d(x, y) : y \in A^c\}) \wedge 1$$

These are continuous functions with $0 \leq f_n \uparrow 1_A$, so $\|f_n \to f\|_p \to 0$ by dominated convergence. By regularity, this result extends to arbitrary $A \in \mathcal{S}$. By linearity this extends to simple measurable function. Then, for general $f \in L^p$ we may choose some simple measurable functions $f_n \to f$ with $|f_n| \leq |f|$. Since $|f_n - f|^p \leq 2^{p+1} |f|^p$, we get $\|f_n - f\|_p \to 0$ by dominated convergence, and the theorem follows.

■

Theorem 11.3: Egorov's Theorem

Let $f, f_1, f_2, \ldots$ be measurable functions on some finite measure space $(\Omega, \mathcal{A}, \mu)$ such that $f_n \to f$ on Ω. Then, for any $\epsilon > 0$, there exists some $A \in \mathcal{A}$ with $\mu A^c < \epsilon$ such that $f_n \to f$ uniformly on A.

Theorem 11.4: Lusin's Theorem

Let f be a measurable function on some compact metric space S with Borel σ-field $\mathcal{S}$ and a bounded measure μ. Then, there exist some continuous functions $f_1, f_2, \ldots$ on S such that $\mu\{x : f_n(x) \not\to f(x)\} \to 0$.

Definition 11.5: Outer Measure

An *outer measure* on Ω, is a nondecreasing and countable subadditive function $\mu : 2^\Omega \to \overline{\mathbb{R}}_+$ with $\mu\emptyset = 0$. We say a set $A \subseteq \Omega$ is *μ-measurable* if

$$\mu(E) = \mu(E \cap A) + \mu(E \cap A^c), \quad E \subseteq \Omega$$

We will ignore most of these results as we will never hear about this again, but the important result is as follows: Denote the length of an interval $I \subseteq \mathbb{R}$ as $|I|$. Then, there exists a unique measure on $\mathbb{R}$ for which $\lambda(I) = |I|$ for every interval.

From a previous theorem, we also know that we can construct the product measure $\lambda^d = \lambda \otimes \cdots \otimes \lambda$ on $\mathbb{R}^d$ for every $d \in \mathbb{N}$. We call λ^d the *d-dimensional Lebesgue measure*. Note that, by how we defined it, λ^d generalizes the ordinary notion of area when $d = 2$ and volume when $d \geq 3$.

Interestingly, it turns out that the Lebesgue measure λ is the unique measure that is invariant under translation and such that $\lambda((0, 1)) = 1$.

A useful result related to this is that, if we have a measureable space $(S, \mathcal{S})$ and a measure μ on $\mathbb{R}^d \times S$ with σ-finite projection $\nu = \mu((0, 1]^d \times \cdot)$ onto S. Then, μ is invariant under shifts in $\mathbb{R}^d$ iff $\mu = \lambda^d \otimes \nu$, in which case μ remains invariant under arbitrary rigid motions of $\mathbb{R}^d$.

10/8/26 Lecture 12: Signed Measures and the Radon-Nikodym Theorem

Definition 12.1: Bounded Signed Measures

A *bounded signed measure* on a measurable space $(\Omega, \mathcal{A})$ is a bounded function $\nu : \mathcal{A} \to \mathbb{R}$ such that

$$\nu\left(\bigcup_n A_n\right) = \sum_n \nu A_n$$

for disjoint $A_i \in \mathcal{A}$, where this series converges absolutely.

Definition 12.2: Mutually Singular Measures

We say μ and ν on $(\Omega, \mathcal{A})$ are *mutually singular* or *orthogonal*, and write $\mu \perp \nu$, if there exists $A \in \mathcal{A}$ with $\mu A = \nu A^c = 0$.

Example

The Lebesgue measure and the counting measure on the Cantor set are mutually singular.

Proposition 12.3

If α, β are bounded measures, then $\nu = \alpha - \beta$ is a bounded signed measure.

Proof: If $\sum_n x_n$ and $\sum_n y_n$ are absolutely convergent, then $\sum_n (x_n - y_n)$ is absolutely convergent.

Theorem 12.4: Hahn Decomposition

Any bounded signed measure ν can be written uniquely as a difference of two bounded, non-negative, and mutually singular measures ν_+ and ν_-.

Proof: (Existence) Let $c = \sup\{\nu A : A \in \mathcal{A}\}$. If $A, A' \in \mathcal{A}$ with $\nu A \geq c - \epsilon$ and $\nu A' \geq c - \epsilon'$, then

$$\begin{aligned}\nu(A \cup A') &= \nu A + \nu A' - \nu(A \cap A') \\ &\geq (c - \epsilon) + (c - \epsilon') - c = c - \epsilon - \epsilon'\end{aligned}$$

Choosing $A_1, A_2, \ldots \in \mathcal{A}$ with $\nu A_n \geq c - 2^{-n}$, we see that, by the above,

$$\nu\left(\bigcup_{k \geq n} A_k\right) \geq c - \sum_{k \geq n} 2^{-k} = c - 2^{-n+1}, \quad n \in \mathbb{N}$$

Let $A_+ = \bigcap_n \bigcup_{k \geq n} A_k$ and $A_- = A_+^c$. By continuity, $\nu A_+ = c$. Hence, for $B \in \mathcal{A}$,

$$\begin{aligned}\nu B &= \nu A_+ - \nu(A_+ \backslash B) \geq 0, \quad B \subseteq A_+ \\ \nu B &= \nu(A_+ \cup B) - \nu A_+ \leq 0, \quad B \subseteq A_-\end{aligned}$$

as $\nu A_+ = \sup\{\nu A : A \in \mathcal{A}\}$. Then, define ν_+ and ν_- by

$$\nu_+ B = \nu(B \cap A_+) \quad \text{and} \quad \nu_- B = -\nu(B \cap A_-), \quad B \in \mathcal{A}$$

These are both clearly measures, and they are bounded and mutually singular.

(Uniqueness) Assume that $\nu = \mu_+ - \mu_-$ for some positive measures $\mu_+ \perp \mu_-$. Choose $B_+ \in \mathcal{A}$ with $\mu_- B_+ = \mu_+ B_+^c = 0$. Then, we first have that $\nu(A_+ \setminus B_+) \geq 0$, as otherwise it wouldn't be included in A_+, but we also have

$$\nu(A_+ \cap B_+^c) = \mu_+(A_+ \cap B_+^c) - \mu_-(A_+ \cap B_+^c) = -\mu_-(A_+ \cap B_+^c) \leq 0$$

So $\nu(A_+ \setminus B_+) = 0$. By a similarly argument, $\nu(B_+ \setminus A_+) = 0$, so $\nu = 0$ on $A_+ \triangle B_+$. Hence, for any $C \in \mathcal{A}$,

$$\mu_+ C = \mu_+(B_+ \cap C) = \nu(B_+ \cap C) = \nu(A_+ \cap C) = \nu_+ C$$

where the first equality comes from the fact μ_+ is zero on B_+^c, the second equality comes from the fact μ_- is zero on B_+, and the third fact comes from the fact that $\nu(B_+) = \nu(B_+ \cap A_+) = \nu(A_+)$. Thus, $\mu_+ = \nu_+$. Then, as $\mu_- = \mu_+ - \nu = \nu_+ - \nu = \nu_-$, we have the other result.

■

Corollary 12.5: Maximum and Minimum

For any σ-finite measures μ and ν on a common measurable space, there exists a largest measure $\mu \wedge \nu$ bounded by μ and ν (i.e. $\mu \wedge \nu \leq \min\{\mu, \nu\}$) and a smallest measure $\mu \vee \nu$ bounding μ and ν (i.e. $\mu \vee \nu \geq \max\{\mu, \nu\}$). Furthermore,

$$\mu - \mu \wedge \nu \perp \nu - \mu \wedge \nu \quad \text{and} \quad \mu \wedge \nu + \mu \vee \nu = \mu + \nu$$

Proof: We may assume that μ and ν are bounded. If not, note that given a μ-finite cover of X and a ν-finite cover of X, we can intersect them giving us a cover that is both μ and ν finite, in which case we can restrict and add the up the result in the end. Anyways, letting $\rho_+ - \rho_-$ be the Hahn decomposition of $\mu - \nu$, let

$$\mu \wedge \nu = \mu - \rho_+ = \nu - \rho_-$$
$$\mu \vee \nu = \mu + \rho_- = \nu + \rho_+$$

These clearly satisfy most of the desired properties, but we will show minimality and maximality. First, letting A_+ be the proof from above, we see that if $B \subseteq A_+$, then $\mu(B) \geq \nu(B)$, and if $B \subseteq A_+^c$, then $\nu(B) \geq \mu(B)$.

Now, suppose that $\lambda \geq \mu$ and $\lambda \geq \nu$. Then, for any $C \in \mathcal{A}$, we see that

$$\lambda(C) = \lambda(C \cap A_+) + \lambda(C \cap A_+^c) \geq \mu(C \cap A_+) + \nu(C \cap A_+^c) = (\mu \vee \nu)(C)$$

and similarly, if $\lambda \leq \mu$ and $\lambda \leq \nu$, then for any $C \in \mathcal{A}$,

$$\lambda(C) = \lambda(C \cap A_+) + \lambda(C \cap A_+^c) \leq \nu(C \cap A_+) + \mu(C \cap A_+^c) = (\mu \wedge \nu)(C)$$

■

Definition 12.6: Absolute Continuity

For any two measures μ and ν on $(\Omega, \mathcal{A})$, ν is said to be *absolutely continuous with respect to* μ if $\mu A = 0$ implies $\nu A = 0$ for all $A \in \mathcal{A}$, in which case we write $\nu \ll \mu$.

Example

If μ is any measure and $f \geq 0$ is measurable, define ν by $\nu A = \mu(f \cdot 1_A)$. Then, $\nu \ll \mu$. We write $\nu = f \cdot \mu$.

Theorem 12.7: Radon-Nikodym and Lebesgue Decomposition Theorem

For any σ-finite measures μ and ν on Ω, there exist some unique measures $\nu_a \ll \mu$ and $\nu_s \perp \mu$ such that $\nu = \nu_a + \nu_s$. Furthermore, $\nu_a = f \cdot \mu$ (that is, $\nu_a(A) = \mu(f 1_A)$) for some μ-a.e. unique measurable function $f \geq 0$ on Ω.

Proof: We need 2 lemmas.

Lemma 12.8: Closure

Fix two measures μ and ν on Ω and some measurable functions $f_1, f_2, \ldots \geq 0$ on Ω with $f_n \cdot \mu \leq \nu$. Then $f \cdot \mu \leq \nu$, where $f = \sup_n f_n$.

Proof: Assume $f \cdot \mu \leq \nu$ and $g \cdot \mu \leq \nu$, and let $h = f \vee g$. Writing $A = \{f \geq g\}$, we have

$$h \cdot \mu = 1_A h \cdot \mu + 1_{A^c} h \cdot \mu = 1_A f \cdot \mu + 1_{A^c} g \cdot \mu \leq 1_A \cdot \nu + 1_{A^c} \cdot \nu = \nu$$

So we can assume $f_n \uparrow f$. Then, $\nu \geq f_n \cdot \mu \uparrow f \cdot \mu$ by the MCT, so $f \cdot \mu \leq \nu$.

■

10/13/26 Lecture 13: Processes, Distributions, and Independence

Lemma 13.1: Partial Density

Let μ and ν be bounded measures on Ω with $\mu \not\perp \nu$. Then, there exists a measurable function $f \geq 0$ on Ω such that $\mu f > 0$ and $f \cdot \mu \leq \nu$.

Proof: For each $n \in \mathbb{N}$, we introduce the signed measure $\chi_n = \nu - n^{-1}\mu$. By the Hahn decomposition, we can find $A_n^+ \in \mathcal{A}$ with complement A_n^- such that $\pm\chi_n \geq 0$ on $A_n^\pm$. Since the χ_n are non-decreasing, we can assume that $A_1^+ \subseteq A_2^+ \subseteq \cdots$ (we get rid of measure 0 sets). Writing $A = \bigcup_n A_n^+$, and noting that $A^c = \bigcap_n A_n^c \subseteq A_n^-$, we see that

$$\nu A^c \leq \nu A_n^- = \chi_n A_n^- + n^{-1}\mu A_n^+ \leq n^{-1}\mu\Omega \to 0$$

so $\nu A^c = 0$. Since $\mu \not\perp \nu$, we must have $\mu A > 0$. Furthermore, $A_n^+ \uparrow A$ implies that $\mu A_n^+ \uparrow \mu A > 0$, so pick n sufficiently large so that $\mu A_n^+ > 0$. Letting $f = n^{-1}1_{A_n^+}$, we see that $\mu f = n^{-1}\mu A_n^+ > 0$, so

$$f \cdot \mu = n^{-1}1_{A_n} \cdot \mu = 1_{A_n^+} \cdot \nu - 1_{A_n^+} \cdot \chi_n \leq \nu$$

■

Now, we can prove the Radon-Nikodym theorem.

Proof of Theorem 12.7 (Radon-Nikodym): Assume that μ and ν are bounded. Let $\mathcal{C}$ denote the class of nonnegative measurable functions f on Ω with $f \cdot \mu \leq \nu$, and let $c = \sup\{\mu f : f \in \mathcal{C}\}$. Choose $f_1, f_2, \ldots \in \mathcal{C}$ with $\mu f_n \to c$, and let $f = \sup_n f_n$. We see that $\sup_n f_n \in \mathcal{C}$ from last time (Lemma 12.8), and as $\mu f \geq \mu f_n$ and $\mu f_n \to c$, we see that $\mu f \geq c$. Thus, $\mu f = c$

Define $\nu_a = f \cdot \mu$ and $\nu_s = \nu - \nu_a$. Note that $\nu_a \ll \mu$ by definition. If $\nu_s \not\perp \mu$, then by Lemma 13.1, there exists $g \geq 0$ with $\mu g > 0$ and $g \cdot \mu \leq \nu_s$. But then $f + g \in \mathcal{C}$ (as $(g + f) \cdot \mu \leq \nu$) and $\mu(f + g) > c$, which contradicts the definition of c. Thus, $\nu_s \perp \mu$.

To prove the uniqueness, suppose that $\nu = \nu_a' + \nu_s'$. Now, pick $A, B \in \mathcal{A}$ so that $\nu_s(A) = \mu(A^c) = \nu_s'(B) = \mu(B^c) = 0$. Then, we see that $\nu_a(A^c) = \nu_a'(B^c) = 0$, so

$$\nu_s(A \cap B) = \nu_s'(A \cap B) = \nu_a(A^c \cup B^c) = \nu_a'(A^c \cup B^c) = 0$$

Thus, we see that

$$\nu_a = 1_{A\cap B} \cdot \nu_a = 1_{A\cap B} \cdot \nu = 1_{A\cap B} \cdot \nu_a' = \nu_a'$$

and

$$\nu_s = \nu - \nu_a = \nu - \nu_a' = \nu_s'$$

which was to be shown. Then, to show that f is a.e. unique, suppose that $\nu_a = g \cdot \mu$ for some measurable $g \geq 0$. Then, we see that, letting $h = f - g$, we see that $\int_A h d\mu = 0$ for all A, which implies that $h = 0$ μ-a.e., so $f = g$ μ-a.e.

Now, to extend to the σ-finite case, write $X = \sqcup X_i$ with X_i having finite measure with respect to both μ and ν, and then use the above theorem on the restricted measure $\mu|_{X_i}$, $\nu|_{X_i}$. Then, piece the results together. ■

Processes, Distributions, and Independence

Example: Fair Coin Tosses

Let $\Omega = \{0,1\}^{\mathbb{N}}$, and let $\mathcal{A}$ be the product σ-field, where $\{0,1\}$ has the discrete σ-field. Let $\mathbb{P}$ be the probability measure such that

$$\mathbb{P}\{\omega \in \Omega : (\omega_1, \dots, \omega_n) = (x_1, \dots, x_n)\} = 2^{-n}$$

for all $n \in \mathbb{N}$. We call this the "fair coin toss measure". We want to show that

$$\mathbb{P}\left\{\omega \in \Omega : \lim_{n\to\infty} \frac{1}{n}(\omega_1 + \cdots + \omega_n) = \frac{1}{2}\right\} = 1$$

By the definition of a limit, this is equivalent to showing that for all $\epsilon > 0$,

$$\mathbb{P}\left\{\omega \in \Omega : \left|\frac{1}{n}(\omega_1 + \cdots + \omega_n) - \frac{1}{2}\right| > \epsilon \text{ for infinitely many } n\right\} = 0$$

This is also equivalent to showing that

$$\mathbb{P}\left\{\omega \in \Omega : \exists N_0 \text{ s.t. } \forall n \geq N_0,\ \left|\frac{1}{n}(\omega_1 + \cdots + \omega_n) - \frac{1}{2}\right| \leq \epsilon\right\} = 1$$

Definition 13.2: Probability Spaces

Fix a probability space $(\Omega, \mathcal{A}, P)$, where P is a *probability measure*; that is, a measure with total mass 1. We call the sets $A \in \mathcal{A}$ *events*, and $PA = P(A)$ is the *probability* of A.

Definition 13.3: Infinitely Often and Ultimately

Given a sequence of events $A_1, A_2, \dots \in \mathcal{A}$, define

$$\{A_n \text{ i.o.}\} = \left\{\omega \in \Omega : \sum_n 1_{A_n}(\omega) = \infty\right\} = \bigcap_n \bigcup_{k\geq n} A_k$$

$$\{A_n \text{ ult.}\} = \left\{\omega \in \Omega : \sum_n 1_{A_n^c}(\omega) < \infty\right\} = \bigcup_n \bigcap_{k\geq n} A_k$$

which we call "A_n *infinitely often*" and "A_n *ultimately*".

Note that

$$1\{A_n \text{ i.o.}\} = \limsup_{n\to\infty} 1_{A_n} \quad \text{and} \quad 1\{A_n \text{ ult.}\} = \liminf_{n\to\infty} 1_{A_n}$$

and by Fatou's lemma,

$$P\{A_n \text{ i.o.}\} \geq \limsup_{n\to\infty} PA_n \quad \text{and} \quad P\{A_n \text{ ult.}\} \leq \liminf_{n\to\infty} PA_n$$

Furthermore,

$$P\{A_n \text{ i.o.}\} = \lim_{n\to\infty} P\left(\bigcup_{k\geq n} A_k\right) \leq \lim_{n\to\infty} \sum_{k\geq n} PA_k$$

which gives us the following result.

Lemma 13.4: First Borel-Cantelli Lemma

If $\sum_n PA_n < \infty$, then $P\{A_n \text{ i.o.}\} = 0$.

Definition 13.5: Random Elements and Distributions

Any measurable mapping ξ of Ω into some measurable space $(S, \mathcal{S})$ is called a *random element* in S. If $B \in \mathcal{S}$, then $\{\xi \in B\} = \xi^{-1}B \in \mathcal{A}$, so we can define the push-forward probability measure

$$P\{\xi \in B\} = P(\xi^{-1}B) = (P \circ \xi^{-1})B, \quad B \in \mathcal{S}$$

The probability measure $\mathcal{L}(\xi) = P \circ \xi^{-1}$ on S induced by ξ is called the *distribution* or *law* of ξ.

Definition 13.6: Types of Random Elements

A random element in S is called

- a *random variable* if $S = \mathbb{R}$,
- a *random vector* if $S = \mathbb{R}^d$,
- a *random sequence* if $S = \mathbb{R}^\infty$,
- a *random* or *stochastic process* when S is a function space,
- a *random measure* or *random set* if S is a class of measures or sets.

We assume that a metric or topological space S is endowed with the Borel σ-field $\mathcal{B}(S)$ unless otherwise specified. For any separable metric space S, $\xi = (\xi_1, \xi_2, \ldots)$ is a random element in S^∞ if, and only if, $\xi_1, \xi_2, \ldots$ are random elements in S.

Example: Random Measures

For random measures, we need a σ-algebra on the space of measures. Here's one way. Let S be the set of finite measures on some space $(X, \mathcal{F})$. For every $B \in \mathcal{F}$, we want the function $f_B : S \to \mathbb{R}$ given by $f(\mu) = \mu(B)$ to be measurable. Then, define

$$\mathcal{A} = \sigma\{f_B : B \in \mathcal{F}\}$$

10/15/26 Lecture 14: Random Processes and Moments

If $(S, \mathcal{S})$ is a measurable space, then any $A \subseteq S$ (we don't need $A \in \mathcal{S}$!) becomes a measurable space with the σ-field $A \cap \mathcal{S} = \{A \cap B : B \in \mathcal{S}\}$. Then, any random element of $(A, A \cap \mathcal{S})$ can be thought of as a random element in S.

Conversely, if ξ is a random element in S such that $\xi \in A$ a.s. (that is, $P(\xi^{-1}(A^c)) = 0$) for some $A \in \mathcal{S}$, then $\xi = \eta$ almost surely for some random element η in A.

Definition 14.1: Evaluation Maps

Fixing a measurable space $(S, \mathcal{S})$ and an abstract index set T, write S^T for the class of functions $f : T \to S$, and let $\mathcal{S}^T$ denote the σ-field in S^T generated by all *evaluation maps* $\pi_t : S^T \to S$, $t \in T$, given by

$$\pi_t f = f(t)$$

If $X : \Omega \to \mathcal{U} \subseteq S^T$, then clearly $X_t = \pi_t \circ X$ maps Ω into S. Thus, X may be regarded as a function $X(\omega, t) = X_t(\omega)$ from $\Omega \times T$ into S. That is,

$$X(\omega, t) = (X(\omega))(t)$$

Example

Let $S = \mathbb{R}$ and $T = [0, 1]$, so $S^T = \mathbb{R}^{[0,1]}$ is the set of all functions $f : [0, 1] \to \mathbb{R}$. Let $\mathcal{U} = C([0, 1], \mathbb{R})$ be the continuous functions from $[0, 1]$ to $\mathbb{R}$. Then, a map $X : \Omega \to C([0, 1], \mathbb{R})$ can be thought of as a function $X : \Omega \times [0, 1] \to \mathbb{R}$ by

$$X(\omega, t) = (X(\omega))(t)$$

Lemma 14.2

Fix a measurable space $(S, \mathcal{S})$, an index set T, and a subset $\mathcal{U} \subseteq S^T$. Then, a function $X : \Omega \to \mathcal{U}$ is $\mathcal{U} \cap \mathcal{S}^T$-measurable if, and only if, $X_t : \Omega \to S$ is $\mathcal{S}$-measurable for all $t \in T$.

Definition 14.3: Random Processes

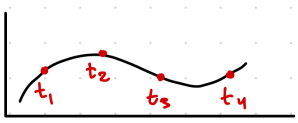
A mapping X as above is called an *S-valued random process on T with paths in $\mathcal{U}$*. By the lemma, we can regard X as a collection of random elements X_t in the *state space* S.



Definition 14.4: Finite-Dimensional Distributions

For any random elements ξ and η in a common measurable space, the equality $\xi \overset{d}{=} \eta$ means that ξ and η have the same distribution, or $\mathcal{L}(\xi) = \mathcal{L}(\eta)$. If X is a random process on some index set T, the associated *finite-dimensional distributions* are given by

$$\mu_{t_1,\dots,t_n} = \mathcal{L}(X_{t_1}, \dots, X_{t_n}), \quad t_1, \dots, t_n \in T,\ n \in \mathbb{N}$$



Proposition 14.5: Finite-Dimensional Distributions

Fix S, T, and $\mathcal{U} \subseteq S^T$, and let X and Y be processes on T with paths in $\mathcal{U}$. Then, $X \overset{d}{=} Y$ if, and only if,

$$(X_{t_1}, \dots, X_{t_n}) \overset{d}{=} (Y_{t_1}, \dots, Y_{t_n}), \quad t_1, \dots, t_n \in T,\ n \in \mathbb{N}$$

Proof:

Definition 14.6: Distribution Functions

For any random vector $\xi = (\xi_1, \dots, \xi_d)$ in $\mathbb{R}^d$, we define the associated *distribution function* F by

$$F(x_1, \dots, x_d) = P\left(\bigcap_{k \le d} \{\xi_k \le x_k\}\right), \quad x_1, \dots, x_d \in \mathbb{R}$$

Lemma 14.7: Distribution Functions

Let ξ and η be random vectors in $\mathbb{R}^d$ with distribution functions F and G. Then, $\xi \overset{d}{=} \eta$ if, and only if, $F = G$.

Proof: This follows from the π-λ theorem.

Definition 14.8: Expected Value

The *expected value*, *expectation*, or *mean* of a random variable ξ is defined as

$$\mathbb{E}[\xi] = \int_\Omega \xi dP = \int_{\mathbb{R}} x(P \circ \xi^{-1})(dx)$$

That is, we can either integrate ξ over Ω, or integrate the possible images with respect to the push-forward measure.

For any random element ξ in some measurable space S and an arbitrary measurable function $f : S \to \mathbb{R}$,

$$\begin{aligned}\mathbb{E}[f(\xi)] &= \int_\Omega f(\xi) dP = \int_S f(s)(P \circ \xi^{-1})(ds) \\ &= \int_{\mathbb{R}} x(P \circ (f \circ \xi)^{-1})(dx)\end{aligned}$$

provided one of these three integrals exists. Integrals over a measurable subset $A \subseteq \Omega$ are often denoted by

$$\mathbb{E}[\xi; A] = \mathbb{E}[\xi 1_A] = \int_A \xi dP, \quad A \in \mathcal{A}$$

Definition 14.9: Absolute Moments

For any random variable ξ and constant $p > 0$, the integral $\mathbb{E}\,|\xi|^p = \|\xi\|_p^p$ is called the *pth absolute moment* of ξ.

By Hölder's inequality (or Jensen's, using the convexity of x^p), we see that $\|\xi\|_p \leq \|\xi\|_q$ for $p \leq q$, so the corresponding L^p-spaces are non-increasing in p.

Here is why. We want to show that $(\mathbb{E}\,|\xi|^p)^{1/p} \leq (\mathbb{E}\,|\xi|^q)^{1/q}$. Let $\eta = |\xi|^p$. So, we want to show that $\mathbb{E}[|\eta|] \leq \mathbb{E}[|\eta|^{q/p}]^{p/q}$, or $\mathbb{E}[|\eta|] \leq \mathbb{E}[|\eta|^r]^{1/r}$ where $r = q/p \geq 1$. Then, by Hölder's,

$$\mathbb{E}[|\eta|] = \mathbb{E}[|\eta| \cdot 1] \leq \|\eta\|_r \|1\|_s = \|\eta\|_r = \mathbb{E}[|\eta|^r]^{1/r}$$

where $\|1\|_s = 1$ as we have a probability space, and where we use $p \leq q$ to ensure $r = \frac{q}{p} \geq 1$, so that there exists s with $\frac{1}{r} + \frac{1}{s} = 1$.

Definition 14.10: Moments

If $\xi \in L^p$ and either $p \in \mathbb{N}$ or $\xi \geq 0$, define the *pth moment* of ξ as $\mathbb{E}[\xi^p]$.

Note that we require $\xi \geq 0$ or $p \in \mathbb{N}$ to ensure we don't raise negative numbers to rational powers.

Lemma 14.11: Moments and Tails

For any random variable $\xi \geq 0$,

$$\mathbb{E}[\xi^p] = p \int_0^\infty P\{\xi > t\} t^{p-1} dt = p \int_0^\infty P\{\xi \geq t\} t^{p-1} dt, \quad p > 0$$

Proof: We see that

$$\begin{aligned}
\mathbb{E}[\xi^p] &= \int_\Omega \xi^p P(d\omega) \\
&= \int_\Omega \left(\int_0^{\xi(\omega)} pt^{p-1} dt \right) P(d\omega) \\
&= \int_\Omega \left(\int_0^{\infty} pt^{p-1} 1\{t \le \xi(\omega)\} dt \right) P(d\omega) \\
&= \int_0^\infty \int_\Omega pt^{p-1} 1\{t \le \xi(\omega)\} P(d\omega) dt \\
&= \int_0^\infty pt^{p-1} P(\xi \ge t) dt
\end{aligned}$$

where we swap the integrals using Tonelli and the fact th integrant is nonnegative. The $>$ follows as the difference of these sets is measure zero. The $\zeta > t$ version follows as $P(\xi \ge t) = P(\xi > t)$ except at the countably many t with $P(\xi = t) > 0$

■

10/20/26 Lecture 15: Jensen's Inequality, Covariance, and Independence

Definition 15.1: Integrable Random Vectors and Processes

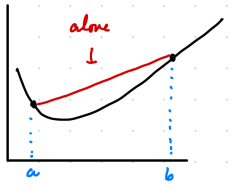
A random vector $\xi = (\xi_1, \dots, \xi_d)$ or process $X = (X_t)$ is said to be *integrable* if integrability holds for every ξ_k or X_t. Then, we write

$$\mathbb{E}\xi = (\mathbb{E}\xi_1, \dots, \mathbb{E}\xi_d) \quad \text{or} \quad \mathbb{E}X = (\mathbb{E}X_t)$$

Definition 15.2: Convex Functions

A function $f : \mathbb{R}^d \to \mathbb{R}$ is said to be *convex* if

$$f(px + (1-p)y) \le pf(x) + (1-p)f(y), \quad x, y \in \mathbb{R}^d,\ p \in [0,1]$$



Lemma 15.3: Jensen's Inequality

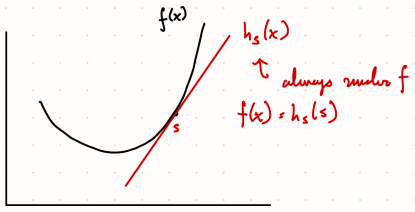
For any integrable random vector ξ in $\mathbb{R}^d$ and convex $f : \mathbb{R}^d \to \mathbb{R}$, we have

$$\mathbb{E}[f(\xi)] \ge f(\mathbb{E}[\xi])$$

Proof: Convexity implies the existence, for every $s \in \mathbb{R}^d$, of a function $h_s(x) = a^T x + b$ with $f \ge h_s$ and $f(s) = h_s(s)$. Then, if $s = \mathbb{E}(\xi)$,

$$\mathbb{E}f(\xi) \ge \mathbb{E}h_s(\xi) = h_s(\mathbb{E}\xi) = f(\mathbb{E}\xi)$$

where the second equality comes from linearity.



■

Definition 15.4: Covariance

The *covariance* of two random variables $\xi, \eta \in L^2$ is given by

$$\operatorname{cov}(\xi, \eta) = \mathbb{E}[(\xi - \mathbb{E}\xi)(\eta - \mathbb{E}\eta)] = \mathbb{E}[\xi\eta] - \mathbb{E}[\xi]\mathbb{E}[\eta]$$

Covariance is *bilinear*, so

$$\operatorname{cov}\left(\sum_{j\le m} a_j\xi_j, \sum_{k\le n} b_k\eta_k\right) = \sum_{j\le m}\sum_{k\le n} a_j b_k \operatorname{cov}(\xi_j, \eta_k)$$

Definition 15.5: Variance

For $\xi \in L^2$, we define the *variance*

$$\operatorname{var}(\xi) = \operatorname{cov}(\xi, \xi) = \mathbb{E}((\xi - \mathbb{E}\xi)^2) = \mathbb{E}\xi^2 - (\mathbb{E}\xi)^2$$

By Cauchy-Buniakovsky,

$$|\operatorname{cov}(\xi, \eta)| \le [\operatorname{var}(\xi)\operatorname{var}(\eta)]^{1/2}$$

which gives that the correlation

$$\operatorname{Corr}(\xi, \eta) = \frac{\operatorname{cov}(\xi, \eta)}{\sqrt{\operatorname{var}(\xi)\operatorname{var}(\eta)}} \in [-1, 1]$$

Definition 15.6: Uncorrelated Random Variables

We say ξ and η are *uncorrelated* if $\operatorname{cov}(\xi, \eta) = 0$.

Definition 15.7: Covariance Functions

For any collection of random variables $\xi_t \in L^2$, $t \in T$, the associated *covariance function* $\rho_{s,t} = \operatorname{cov}(\xi_s, \xi_t)$, $s, t \in T$, is *non-negative definite*, in the sense that

$$\sum_{i,j} a_i a_j \rho_{t_i,t_j} \ge 0 \quad \text{for any } n \in \mathbb{N},\ t_1, \dots, t_n \in T, \text{ and } a_1, \dots, a_n \in \mathbb{R}$$

This comes from

$$\sum_{i,j} a_i a_j \rho_{t_i,t_j} = \sum_{i,j} a_i a_j \operatorname{cov}(\xi_{t_i}, \xi_{t_j}) = \operatorname{var}\left(\sum_i a_i \xi_{t_i}\right) \ge 0$$

where the last inequality is by Jensen's inequality.

Definition 15.8: Mutually Independent Events

Events $A_t \in \mathcal{A}$, $t \in T$ are said to be *mutually independent* if, for any distinct $t_1, \dots, t_n \in T$,

$$P\left(\bigcap_{k\le n} A_{t_k}\right) = \prod_{k\le n} P(A_{t_k})$$

Definition 15.9: Independent Families

More generally, the families $\mathcal{C}_t \subseteq \mathcal{A}$, $t \in T$ are *independent* if independence holds between events A_t for arbitrary $A_t \in \mathcal{C}_t$, $t \in T$.

Definition 15.10: Independent Random Elements

Random elements ξ_t, $t \in T$ are *independent* if independence holds between the σ-fields $\sigma(\xi_t)$, $t \in T$.

Pairwise independence is denoted $A \perp\!\!\!\perp B$, $\xi \perp\!\!\!\perp \eta$, or $\mathcal{B} \perp\!\!\!\perp \mathcal{C}$, respectively.

Lemma 15.11: Extension

If the π-systems $\mathcal{C}_t$, $t \in T$ are independent, so are the σ-fields $\mathcal{F}_t = \sigma(\mathcal{C}_t)$, $t \in T$.

Proof: First, assume that $\mathcal{C}_t$ is nonempty for any t, and fix arbitrary indices $t_1, \dots, t_n \in T$. Notice that for a collection $A_{t_k} \in \mathcal{C}_{t_k}$, independence holds by assumption.

Now, fixing $A_{t_2}, \dots, A_{t_n}$, let $\mathcal{D}$ be the collection of sets $A \in \mathcal{A}$ for which independence holds in the collection $A, A_{t_2}, \dots, A_{t_n}$. This is a λ-system that contains $\mathcal{C}_{t_1}$, so $\mathcal{F}_{t_1} \subseteq \mathcal{D}$ by the π-λ theorem.

Thus, independence holds for arbitrary $A_{t_1} \in \mathcal{F}_{t_1}$ and $A_{t_k} \in \mathcal{C}_{t_k}$ for $k = 2, \dots, n$.

Now, apply the same argument fixing instead $A_{t_1} \in \mathcal{F}_{t_1}$ and $A_{t_k} \in \mathcal{C}_{t_k}$ for $k = 3, \dots, n$ to show that $\mathcal{F}_{t_2} \subseteq \mathcal{D}$, and proceed inductively.

■

Let $\mathcal{F} \vee \mathcal{G} = \sigma\{\mathcal{F}, \mathcal{G}\}$ and $\mathcal{F}_S = \bigvee_{t \in S} \mathcal{F}_t = \sigma\{\mathcal{F}_t : t \in S\}$.

Corollary 15.12: Grouping

Let $\mathcal{F}_t$, $t \in T$ be independent σ-fields, and let $\mathcal{T}$ be a disjoint partition of T. Then, the σ-fields $\mathcal{F}_S = \bigvee_{t \in S} \mathcal{F}_t$, $S \in \mathcal{T}$ are again independent.

Definition 15.13: Separating Classes

Given any set T, we say a class $\mathcal{T} \subseteq 2^T$ is *separating* if, for any $s \neq t$ in T, there exists $S \in \mathcal{T}$ such that exactly one of s and t lies in S.

Lemma 15.14: Pairwise Independence

i) The σ-fields $\mathcal{F}_1, \mathcal{F}_2, \dots$ are independent iff $\bigvee_{k \leq n} \mathcal{F}_k \perp\!\!\!\perp \mathcal{F}_{n+1}$ for all n.

ii) The σ-fields $\mathcal{F}_t$, $t \in T$ are independent iff $\mathcal{F}_S \perp\!\!\!\perp \mathcal{F}_{S^c}$ for all sets S in some separating class $\mathcal{T} \subseteq 2^T$.

Proof: First, the necessary conditions hold by the corollary.

We will show the sufficient condition for (ii), as (i) follows from a similar construction. We want, for any finite $S \subseteq T$, the σ-fields $\mathcal{F}_s$, $s \in S$ are independent.

Let $|S| = n+1$ denote the cardinality of S and assume it holds for all $|S| \leq n$. Choose a separating $U \in \mathcal{T}$ such that $S' = S \cap U$ and $S'' = S \setminus U$ are nonempty. Then, since $\mathcal{F}_{S'} \perp\!\!\!\perp \mathcal{F}_{S''}$, for any sets $A_s \in \mathcal{F}_s$, $s \in S$,

$$P \bigcap_{s \in S} A_s = \left(P \bigcap_{s \in S'} A_s \right) \left(P \bigcap_{s \in S''} A_s \right) = \prod_{s \in S} P A_s$$

where the last relation holds by the induction hypothesis as $|S'|, |S''| < |S| = n+1$.

We need to check this is true for $|S| = 1$, but this is trivial.

■

10/22/26 Lecture 16: Zero-One Laws

Definition 16.1: P-Trivial σ-Fields and Degenerate Random Elements

A σ-field $\mathcal{F}$ is said to be *P-trivial* if $PA = 0$ or 1 for every $A \in \mathcal{F}$. A random element is *a.s. degenerate* if its distribution is a degenerate probability measure; that is,

$$P\{\xi \in A\} = \begin{cases} 1, & x \in A \\ 0, & \text{else} \end{cases}$$

for some $x \in S$. In other words, the distribution of ξ is δ_x.

Lemma 16.2: Triviality and Degeneracy

A σ-field $\mathcal{F}$ is P-trivial iff $\mathcal{F} \perp\!\!\!\perp \mathcal{F}$. In that case, any $\mathcal{F}$-measurable random element ξ taking values in a separable metric space is a.s. degenerate.

Proof: If $\mathcal{F} \perp\!\!\!\perp \mathcal{F}$, then for $A \in \mathcal{F}$, $PA = P(A \cap A) = (PA)^2$, so $PA = 0$ or $PA = 1$.

Conversely, if $\mathcal{F}$ is P-trivial, then for $A, B \in \mathcal{F}$, $P(A \cap B) = \min\{PA, PB\} = PA \cdot PB$, so $\mathcal{F} \perp\!\!\!\perp \mathcal{F}$ (if $PA = PB = 1$, then $P(A \cap B) = 1$, as otherwise $P(A \cup B) > 1$).

Now, assume $\mathcal{F}$ is P-trivial and let ξ be a random element in a separable metric space. For each n, partition S into countably many disjoint Borel sets B_{nj} of diameter $< n^{-1}$.

We can do this by taking a countable dense set $D = \{x_1, x_2, \ldots\}$ and writing, for $\epsilon < 1/(2n)$,

$$B_{n1} = B_\epsilon(x_1),\ B_{n2} = B_\epsilon(x_2) \setminus B_{n1},\ \ldots,\ B_{nj} = B_\epsilon(x_j) \setminus \bigcup_{i=1}^{j-1} B_{ni}$$

This gives us the desired construction.

Since $P\{\xi \in B_{nj}\} = 0$ or 1 and the B_{nj} are disjoint, we have $\xi \in B_{nj}$ a.s. for exactly one $j = j_n$. Then, we have

$$\xi \in \bigcap_{n=1}^{\infty} B_{nj_n}$$

which is a set of zero diameter, so it must be a single point. Thus, $\xi = x$ a.s., so it is a.s. degenerate. ■

Lemma 16.3: Product Measures

Let $\xi_1, \ldots, \xi_n$ be random elements in measurable spaces $S_1, \ldots, S_n$ with distributions $\mu_1, \ldots, \mu_n$. Then, the ξ_k are independent iff $\xi = (\xi_1, \ldots, \xi_n)$ has distribution $\mu_1 \otimes \cdots \otimes \mu_n$.

Proof: Assuming independence, we get for $B = B_1 \times \cdots \times B_n$,

$$P\{\xi \in B\} = \prod_{k\le n} P\{\xi_k \in B_k\} = \prod_{k\le n} \mu_k B_k = \bigotimes_{k\le n} \mu_k B$$

by definition of product measure. Then, use π-λ to extend. The opposite direction uses the same equality above.

■

Lemma 16.4: Conditioning

Let ξ and η be independent random elements in some measurable spaces S and T, and let the function $f : S \times T \to \mathbb{R}$ be measurable with $\mathbb{E}(\mathbb{E}\,|f(s,\eta)|)_{s=\xi} < \infty$. Then,

$$\mathbb{E}f(\xi,\eta) = \mathbb{E}(\mathbb{E}f(s,\eta))_{s=\xi}$$

That is, for each $s \in S$, let $g(s) = \mathbb{E}[f(s,\eta)]$. Then, $\mathbb{E}[f(\xi,\eta)] = \mathbb{E}[g(\xi)]$.

Proof: Let μ and ν denote the distributions of ξ and η, respectively. Assuming $f \ge 0$, and writing $g(s) = \mathbb{E}[F(s,\eta)]$, we see that by Fubini's theorem and the properties of the product measure,

$$\begin{aligned}
\mathbb{E}f(\xi,\eta) &= \int f(s,t)(\mu\otimes\nu)(ds,dt) \\
&= \int \mu(ds) \int f(s,t)\nu(dt) \\
&= \int g(s)\mu(ds) \\
&= \mathbb{E}[g(\xi)]
\end{aligned}$$

For general f, this applies to the function $|f|$, and so $\mathbb{E}[|f(\xi,\eta)|] < \infty$, so the desired relation holds.

■

In particular, for independent random variables $\xi_1, \ldots, \xi_n$,

$$\mathbb{E}\left(\prod_k \xi_k\right) = \prod_k \mathbb{E}\xi_k \quad \text{and} \quad \operatorname{var}\sum_k \xi_k = \sum_k \operatorname{var}\xi_k$$

whenever the expressions on the right exist. We then get that if ξ_1, ξ_2 are independent, $\operatorname{cov}(\xi_1,\xi_2) = 0$.

Definition 16.5: Tail σ-Fields

Given any sequence of σ-fields $\mathcal{F}_1, \mathcal{F}_2, \ldots$, introduce the associated *tail σ-field*

$$\mathcal{T} = \bigcap_n \bigvee_{k\ge n} \mathcal{F}_k = \bigcap_n \sigma\{\mathcal{F}_k \,:\, k \ge n\}$$

Theorem 16.6: Kolmogorov's 0-1 Law

Let $\mathcal{F}_1, \mathcal{F}_2, \ldots$ be independent σ-fields. Then, the tail σ-field $\mathcal{T} = \bigcap_n \bigvee_{k \geq n} \mathcal{F}_k$ is P-trivial.

Proof: For each $n \in \mathbb{N}$, define $\mathcal{T}_n = \bigvee_{k>n} \mathcal{F}_k$, and note that $\mathcal{F}_1, \ldots, \mathcal{F}_n, \mathcal{T}_n$ are independent. Hence, so are the σ-fields $\mathcal{F}_1, \ldots, \mathcal{F}_n, \mathcal{T}$ (as $\mathcal{T} \subseteq \mathcal{T}_n$), so then also $\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{T}$ by induction. Then, $\mathcal{T}_0 \perp\!\!\!\perp \mathcal{T}$, so $\mathcal{T} \perp\!\!\!\perp \mathcal{T}$, as $\mathcal{T} \subseteq \mathcal{T}_0$. Thus, $\mathcal{T}$ is P-trivial.

■

Corollary 16.7: Sums and Averages

Let $\xi_1, \xi_2, \ldots$ be independent random variables, and let $S_n = \xi_1 + \cdots + \xi_n$. Then, each of the sequences (S_n) and (S_n/n) is either a.s. convergent or a.s. divergent. For the latter sequence, the possible limit is a.s. degenerate.

Proof:
$\{S_n \text{ converges}\}$ and $\{S_n/n \text{ converges}\}$ are measurable with respect to the tail σ-field.

Now, $\{\lim S_n = x\}$ is not in the tail σ-field, but $\{\lim S_n/n = x\}$ is in the tail σ-field, so $\lim S_n/n$ must be a.s. degenerate.

■

Definition 16.8: Finite Permutations

A *finite permutation* of $\mathbb{N}$ is a bijective map $p : \mathbb{N} \to \mathbb{N}$ such that $p_n = n$ for all but finitely many n.

For any space S, a finite permutation p of $\mathbb{N}$ induces a permutation T_p on S^∞ given by

$$T_p(s) = s \circ p = (s_{p_1}, s_{p_2}, \ldots), \quad s = (s_1, s_2, \ldots) \in S^\infty$$

Definition 16.9: Symmetric Sets

A set $I \subseteq S^\infty$ is said to be *symmetric* (under finite permutations) if

$$T_p^{-1}(I) \equiv \{s \in S^\infty \,:\, s \circ p \in I\} = I$$

for every finite permutation p of $\mathbb{N}$.

Definition 16.10: Permutation Invariant σ-Field

If $(S, \mathcal{S})$ is a measurable space, the symmetric sets $I \in \mathcal{S}^\infty$ form a sub-σ-field $\mathcal{I} \subseteq \mathcal{S}^\infty$, called the *permutation invariant σ-field* in S^∞.

Notation: A sequence of random elements that are independent with identical distributions is said to be *i.i.d.*

Theorem 16.11: Hewitt-Savage 0-1 Law

Let ξ be an infinite sequence of i.i.d. random elements in some measurable space $(S, \mathcal{S})$, and let $\mathcal{I}$ denote the permutation invariant σ-field in S^∞. Then, the σ-field $\xi^{-1}\mathcal{I}$ is P-trivial.

10/29/26 Lecture 17: Hewitt-Savage, Borel-Cantelli, and Convergence in Probability

Theorem 17.1: Hewitt-Savage 0-1 Law

If ξ is an infinite sequence of i.i.d. random variables in $(S, \mathcal{S})$, and $\mathcal{I}$ is the permutation invariant σ-field in $\mathcal{S}^\infty$, then the σ-field $\xi^{-1}(\mathcal{I})$ is P-trivial.

Recall

$$A \triangle B = (A \setminus B) \cup (B \setminus A)$$

and

$$P(A \triangle B) = P(A^c \triangle B^c) = \mathbb{E}\left|1_A - 1_B\right|, \quad A, B \in \mathcal{A}$$

We prove a quick lemma.

Lemma 17.2: Approximation

Given σ-fields $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots$ and a set $A \in \bigvee_n \mathcal{F}_n$, there exist $A_1, A_2, \ldots \in \bigcup_n \mathcal{F}_n$ with $P(A \triangle A_n) \to 0$.

Proof: Let $\mathcal{C} = \bigcup_n \mathcal{F}_n$ and let $\mathcal{D}$ denote the class of sets $A \in \bigvee_n \mathcal{F}_n$ with the property. $\mathcal{C}$ is a π-system and $\mathcal{D}$ is a λ-system, so $\bigvee_n \mathcal{F}_n \subseteq \mathcal{D}$.

Proving that $\mathcal{C}$ is a π-system is easy, but proving $\mathcal{D}$ is a λ-system can be hard. For complements, $A \subseteq B$ with $P(A \triangle A_n) \to 0$ and $P(B \triangle B_n) \to 0$,

$$P((B \setminus A) \triangle (B_n \setminus A_n)) \leq P(B \triangle B_n) + P(A \triangle A_n)$$

so we get the desired result.

Now, suppose that $A^j \in \mathcal{D}$ with $A^j \uparrow A$. Let $\epsilon > 0$. Since $A \triangle A^j \downarrow \emptyset$, we see there exists a j such that $P(A \triangle A^j) < \epsilon/2$. Also, since $A^j \in \mathcal{D}$, we can find a $C \in \mathcal{C}$ with $P(A^j \triangle A) < \epsilon/2$, so

$$P(A \triangle C) \leq P(A \triangle A^j) + P(A^j \triangle C) < \epsilon$$

Thus, taking $\epsilon = 1/k$ for each k we can find sets $C^k \in \mathcal{C}$ with $P(A \triangle C^k) \to 0$. Thus, $A \in \mathcal{D}$.

Thus, $\mathcal{D}$ is a λ-system.

■

Now, we can prove the Hewitt-Savage 0-1 Law.

Proof of Hewitt-Savage: Let $\mu = \mathcal{L}(\xi)$. Let $\mathcal{F}_n = \mathcal{S}^n \times S^\infty$ and note $\mathcal{I} \subseteq \mathcal{S}^\infty = \bigvee_n \mathcal{F}_n$. For any $I \in \mathcal{I}$, there exist $B_n \in \mathcal{S}^n$ so that $I_n = B_n \times S^\infty$ satisfies $\mu(I \triangle I_n) \to 0$.

Now, let $\tilde{I}_n = S^n \times B_n \times S^\infty$. By symmetry of μ and I, $\mu\tilde{I}_n = \mu I_n \to \mu I$. Then $\mu(I \triangle \tilde{I}_n) = \mu(I \triangle I_n) \to 0$. Then,

$$\mu(I \triangle (I_n \cap \tilde{I}_n)) \leq \mu(I \triangle I_n) + \mu(I \triangle \tilde{I}_n) \to 0$$

Since $I_n \perp\!\!\!\perp \tilde{I}_n$ under μ,

$$\mu I \leftarrow \mu(I_n \cap \tilde{I}_n) = (\mu I_n)(\mu \tilde{I}_n) \to (\mu I)^2$$

so $\mu I = 0$ or 1, so $P(\xi^{-1}(I)) = 0$ or 1.

■

Corollary 17.3: Random Walk

Let $\xi_1, \xi_2, \ldots$ be i.i.d. non-degenerate random variables, and let $S_n = \xi_1 + \cdots + \xi_n$. Then,

(i) $P\{S_n \in B \text{ i.o.}\} = 0$ or 1 for any $B \in \mathcal{B}$.

(ii) $\limsup_n S_n = \infty$ a.s. or $-\infty$ a.s.

(iii) $\limsup_n (\pm S_n) = \infty$ if the ξ_n are symmetric; that is, $\xi_n \overset{d}{=} -\xi_n$.

Proof: **(i)** $\{S_n \in B \text{ i.o.}\}$ is permutation invariant in the sense for HS.

(ii) We know $\limsup_n S_n = c$ a.s. by HS and a.s. degeneracy. Then, as

$$c = \limsup_n S_{n+1} = \limsup_n (S_{n+1} - \xi_1) + \xi_1 = c + \xi_1$$

so if $|c| < \infty$, then $\xi_1 = 0$ a.s., which contradicts non-degeneracy of ξ_1. Thus, $|c| = \infty$.

(iii) We see

$$c = \limsup_n S_n \geq \liminf_n S_n = -\limsup_n (-S_n) = -c$$

so $-c \leq c \in \{\pm\infty\}$, so $c = \infty$.

■

Theorem 17.4: Borel-Cantelli

Let $A_1, A_2, \ldots \in \mathcal{A}$. Then

(i) $\sum_n PA_n < \infty$ implies $P\{A_n \text{ i.o.}\} = 0$.

(ii) If the A_n are independent, then $\sum_n PA_n = \infty \implies P\{A_n \text{ i.o.}\} = 1$

Proof: **(i)**

$$\mathbb{E}\left(\sum_n 1_{A_n}\right) = \sum_n \mathbb{E}1_{A_n} = \sum_n PA_n < \infty$$

so $\sum_n 1_{A_n} < \infty$ almost surely. Thus, $P\{A_n \text{ i.o.}\} = 0$.

(ii) Assume the A_n are independent and $\sum_n PA_n = \infty$. Using that $1 - x \le e^{-x}$ for all x,

$$\begin{aligned} P\left(\bigcup_{k\ge n} A_k\right) &= 1 - P\left(\bigcap_{k\ge n} A_k^c\right) \\ &= 1 - \prod_{k\ge n} PA_k^c \\ &= 1 - \prod_{k\ge n} (1 - PA_k) \\ &\ge 1 - \prod_{k\ge n} e^{-PA_k} \\ &= 1 - e^{-\sum_{k\ge n} PA_k} = 1 - e^{-\infty} = 1 \end{aligned}$$

Hence, as $n \to \infty$,

$$P\{A_n \text{ i.o.}\} = \lim_{n\to\infty} P\left(\bigcup_{k\ge n} A_k\right) = 1$$

■

Example: Independence is Needed

Let $U \sim \text{Unif}(0,1)$ and $A_n = \{U \in (0, 1/n)\}$. Note we see $P(A_n) = 1/n$, so $\sum_n PA_n = \infty$, but $P\{A_n \text{ i.o.}\} = 0$. Thus, independence is important.

Lemma 17.5: Bienaymé-Chebyshev-Paley-Zygmund Inequality

Let ξ be an $\mathbb{R}_+$-valued random variable with $0 < \mathbb{E}\xi < \infty$. Then,

$$\frac{(\mathbb{E}\xi)^2}{\mathbb{E}[\xi^2]} \left[(1-r)_+\right]^2 \le P\{\xi > r\mathbb{E}\xi\} \le \frac{1}{r}, \quad r \ge 0$$

The upper bound is also known as Markov's inequality.

Proof: Assume WLOG $\mathbb{E}\xi = 1$ (we can redefine $\eta = \xi/\mathbb{E}\xi$).

Since $r1\{\xi \ge r\} \le \xi$, we see that $rP\{\xi \ge r\} \le \mathbb{E}[\xi]$, so $P\{\xi \ge r\} \le \frac{\mathbb{E}[\xi]}{r}$, and just redefine r.

To get the lower bound, note

$$\begin{aligned} 1 = \mathbb{E}[\xi] &= \mathbb{E}[\xi 1\{\xi \le r\}] + \mathbb{E}[\xi 1\{\xi > r\}] \\ &\le r + \mathbb{E}[\xi^2]^{1/2} P\{\xi > r\}^{1/2} \end{aligned}$$

so for $r' := r\mathbb{E}[\xi]$,

$$\begin{aligned}\mathbb{E}[\xi] &\le r\mathbb{E}[\xi] + \mathbb{E}[\xi^2]^{1/2} P\{\xi > r\mathbb{E}[\xi]\}^{1/2} \\ \mathbb{E}[\xi](1-r) &\le \mathbb{E}[\xi^2]^{1/2} P\{\xi > r\mathbb{E}\xi\}^{1/2} \\ \frac{(\mathbb{E}[\xi])^2}{\mathbb{E}[\xi^2]} [(1-r)_+]^2 &\le P\{\xi > r\mathbb{E}\xi\}\end{aligned}$$

■

Definition 17.6: Convergence in Probability

Given random elements ξ and $\xi_1, \xi_2, \dots$ in a metric space (S, ρ), say that ξ_n *converges in probability* to ξ, written $\xi_n \xrightarrow{P} \xi$, if

$$\lim_{n\to\infty} P\{\rho(\xi_n, \xi) > \epsilon\} = 0, \quad \epsilon > 0$$

By Chebyshev's inequality, this is equivalent to $\mathbb{E}[\rho(\xi_n, \xi) \wedge 1] \to 0$. This is because, for $\epsilon < 1$

$$P\{\rho(\xi_n, \xi) > \epsilon\} = P\{\rho(\xi_n, \xi) \wedge 1 > \epsilon\} \le \frac{\mathbb{E}[\rho(\xi_n, \xi) \wedge 1]}{\epsilon}$$

aso the LHS goes to zero as $\mathbb{E}[\rho(\xi_n, \xi) \wedge 1] \to 0$. Similarly, because

$$\mathbb{E}[\rho(\xi_n, \xi) \wedge 1] \le \epsilon + P\{\rho(\xi_n, \xi) > \epsilon\}$$

so taking $n \to \infty$, we see that $\lim_{n\to\infty} \mathbb{E}[\rho(\xi_n, \xi) \wedge 1] \le \epsilon$ for all ϵ.

Here is a more useful definition.

Lemma 17.7: Subsequence Criterion

Let $\xi, \xi_1, \xi_2, \dots$ be random elements in a metric space (S, ρ). Then $\xi_n \xrightarrow{P} \xi$ iff every subsequence ξ_{n_m} has a further subsequence $\xi_{n_{m_k}}$ such that $\xi_{n_{m_k}} \to \xi$ a.s. In particular, $\xi_n \to \xi$ a.s. implies $\xi_n \xrightarrow{P} \xi$.

Proof: ($\Rightarrow$) Assume $\xi_n \xrightarrow{P} \xi$ and let ξ_{n_m} be a subsequence. Since $\mathbb{E}[\rho(\xi_{n_m}, \xi) \wedge 1] \to 0$, pick a further subsequence so that $\mathbb{E}[\rho(\xi_{n_{m_k}}, \xi) \wedge 1] < 2^{-k}$. Then

$$\mathbb{E} \sum_k \{\rho(\xi_{n_{m_k}}, \xi) \wedge 1\} \overset{\text{MCT}}{=} \sum_k \mathbb{E}[\rho(\xi_{n_{m_k}}, \xi) \wedge 1] < \infty$$

Then, $\sum_k \rho(\xi_{n_{m_k}}, \xi) \wedge 1$ is finite a.s., so $\rho(\xi_{n_{m_k}}, \xi) \to 0$ a.s.

($\Leftarrow$) Assume the subsequence condition. Now, if $\xi_n \not\xrightarrow{P} \xi$, there exists $\epsilon > 0$ such that $\mathbb{E}[\rho(\xi_{n_m}, \xi) \wedge 1] > \epsilon$ along a subsequence ξ_{n_m}. By hypothesis, we know $\xi_{n_{m_k}} \to \xi$ a.s. for some further subsequence, so by DCT, we know

$$\mathbb{E}[\rho(\xi_{n_{m_k}}, \xi) \wedge 1] \to 0$$

which is a contradiction.

■

Lemma 17.8: Continuous Mapping

For any metric spaces S, T, let $\xi, \xi_1, \xi_2, \ldots$ be random elements in S with $\xi_n \xrightarrow{P} \xi$ and let $f : S \to T$ be measurable and a.s. continuous at ξ. Then, $f(\xi_n) \xrightarrow{P} f(\xi)$.

Proof: Let ξ_{n_m} be a subsequence. Then, find a further subsequence $\xi_{n_{m_k}} \to \xi$ a.s. Then, by continuity,

$$f(\xi_{n_{m_k}}) \to f(\xi) \quad \text{a.s.}$$

Thus, $f(\xi_{n_m})$ has an a.s. convergent subsequence, so $f(\xi_n) \xrightarrow{P} f(\xi)$ by above. ■

Consider a sequence of metric spaces (S_k, ρ_k) and consider the product space $S = \bigtimes_k S_k = S_1 \times S_2 \times \cdots$ with the product topology. It turns out a convenient metrization is given by

$$\rho(x, y) = \sum_k 2^{-k} \{\rho_k(x_k, y_k) \wedge 1\}, \quad x, y \in \bigtimes_k S_k$$

Note for $x, x^{(1)}, x^{(2)}, \ldots \in \bigtimes_k S_k$, $\lim_n \rho(x^{(n)}, x) = 0$ iff $\lim_n \rho_k(x_k^{(n)}, x_k) = 0$ for all k.

If each S_k is separable, recall $\mathcal{B}(S) = \bigotimes_k \mathcal{B}(S_k)$, so a random element in S is simply a sequence of random elements in S_k, $k \in \mathbb{N}$.

Lemma 17.9: Random Sequences

For any separable metric spaces $S_1, S_2, \ldots$, let $\xi = (\xi_1, \xi_2, \ldots)$ and $\xi^n = (\xi_1^n, \xi_2^n, \ldots)$, $n \in \mathbb{N}$ be random elements in $\bigtimes_k S_k$. Then,

$$\xi^n \xrightarrow{P} \xi \quad \text{iff} \quad \xi_k^n \xrightarrow{P} \xi_k \text{ in } S_k \text{ for each } k$$

Corollary 17.10: Elementary Operations

Let $\xi, \xi_1, \xi_2, \ldots$ and $\eta, \eta_1, \eta_2, \ldots$ be random variables with $\xi_n \xrightarrow{P} \xi$ and $\eta_n \xrightarrow{P} \eta$. Then,

(i) $a\xi_n + b\eta_n \xrightarrow{P} a\xi + b\eta$ for all $a, b \in \mathbb{R}$.

(ii) $\xi_n \eta_n \xrightarrow{P} \xi\eta$.

(iii) $\xi_n / \eta_n \xrightarrow{P} \xi/\eta$ whenever $\eta \neq 0$ and $\eta_n \neq 0$ a.s. for all n.

Proof: $(\xi_n, \eta_n) \xrightarrow{P} (\xi, \eta)$, and as the operations are continuous, it follows. ■

11/3/26 Lecture 18: Weak Convergence, Tightness, and Uniform Integrability

Definition 18.1: Cauchy in Probability

Random elements $\xi_1, \xi_2, \ldots$ in a metric space (S, ρ) are *Cauchy in probability* if $\rho(\xi_m, \xi_n) \xrightarrow{P} 0$ as $m, n \to \infty$, in the sense that $\mathbb{E}[\rho(\xi_m, \xi_n) \wedge 1] \to 0$.

Lemma 18.2: Completeness

Let $\xi_1, \xi_2, \ldots$ be random elements in a complete metric space (S, ρ). Then (ξ_n) is Cauchy in probability or a.s. iff $\xi_n \xrightarrow{P} \xi$ or $\xi_n \to \xi$ a.s., respectively, for some random element ξ in S.

Proof: The a.s. case we already proved.

Now, if $\xi_n \xrightarrow{P} \xi$, then as

$$\mathbb{E}[\rho(\xi_m, \xi_n) \wedge 1] \leq \mathbb{E}[\rho(\xi_m, \xi) \wedge 1] + \mathbb{E}[\rho(\xi_n, \xi) \wedge 1] \to 0$$

we see (ξ_n) is Cauchy in P.

Assume the latter. Let

$$n_k = \inf\left\{ n \geq k \,:\, \sup_{m \geq n} \mathbb{E}[\rho(\xi_m, \xi_n) \wedge 1] \leq 2^{-k} \right\}, \quad k \in \mathbb{N}$$

Then, the n_k are finite and

$$\mathbb{E} \sum_k \{\rho(\xi_{n_k}, \xi_{n_{k+1}}) \wedge 1\} \leq \sum_k 2^{-k} < \infty$$

so $\sum_k \rho(\xi_{n_k}, \xi_{n_{k+1}}) < \infty$ a.s.

Thus, we see that (ξ_{n_k}) is a.s. Cauchy, and as (S, ρ) is complete, it converges a.s. to some limit ξ. Thus, as

$$\mathbb{E}[\rho(\xi_m, \xi) \wedge 1] \leq \mathbb{E}[\rho(\xi_m, \xi_{n_k}) \wedge 1] + \mathbb{E}[\rho(\xi_{n_k}, \xi) \wedge 1]$$

letting $m, k \to \infty$, we see that $\mathbb{E}[\rho(\xi_m, \xi) \wedge 1] \to 0$.

■

Definition 18.3: Weak Convergence

Consider probability measures μ and $\mu_1, \mu_2, \ldots$ on a metric space (S, ρ) with Borel σ-field $\mathcal{S}$. Say that μ_n *converges weakly* to μ (written $\mu_n \xrightarrow{w} \mu$) if

$$\mu_n f \to \mu f \quad \text{for any } f \in C_b(S)$$

where $C_b(S)$ is the set of bounded, continuous functions $f : S \to \mathbb{R}$.

We call this "weak" convergence since it is weaker than convergence in total variation, which is good because

convergence in TV is overly strong ($\delta_{x_n} \to \delta_x$ iff $x_n = x$ for all large n). The reason we choose $C_b(S)$ is that

$$x_n \to x \iff f(x_n) \to f(x) \quad \forall f \in C_b(S)$$

so $\delta_{x_n} \xrightarrow{w} \delta_x \iff x_n \to x$.

Definition 18.4: Convergence in Distribution

If ξ and $\xi_1, \xi_2, \ldots$ are random elements in S, say ξ_n *converges in distribution* to ξ (written $\xi_n \xrightarrow{d} \xi$) if $\mathcal{L}(\xi_n) \xrightarrow{w} \mathcal{L}(\xi)$; that is,

$$\mathbb{E}[f(\xi_n)] \to \mathbb{E}[f(\xi)] \quad \forall f \in C_b(S)$$

Lemma 18.5

Let $\xi, \xi_1, \xi_2, \ldots$ be random elements in (S, ρ). If $\xi_n \xrightarrow{P} \xi$, then $\xi_n \xrightarrow{d} \xi$, and the converse is true when ξ is a.s. constant.

Proof: Assume $\xi_n \xrightarrow{P} \xi$. Suppose for some $f \in C_b(S)$, $\mathbb{E}[f(\xi_n)] \not\to \mathbb{E}[f(\xi)]$. Choose a subsequence $N' \subseteq \mathbb{N}$ such that

$$\inf_{n \in N'} |\mathbb{E}[f(\xi_n)] - \mathbb{E}[f(\xi)]| > 0$$

Now, as $\xi_n \xrightarrow{P} \xi$, there exists a subsequence $N'' \subseteq N'$ such that $\xi_n \to \xi$ a.s. along N''. By continuity and DCT, we see $\mathbb{E}[f(\xi_n)] \to \mathbb{E}[f(\xi)]$ along N'', a contradiction.

Now, assume $\xi_n \xrightarrow{d} s \in S$. Since $\rho(x, s) \wedge 1$ is bounded and continuous in x,

$$\mathbb{E}[\rho(\xi_n, s) \wedge 1] \to \mathbb{E}[\rho(s, s) \wedge 1] = 0$$

so $\xi_n \xrightarrow{P} s$.

■

Definition 18.6: Tightness

A family of random vectors ξ_t, $t \in T$ in $\mathbb{R}^d$ is said to be *tight* if

$$\lim_{r \to \infty} \sup_{t \in T} P\{|\xi_t| > r\} = 0$$

For sequences (ξ_n), this is equivalent to

$$\lim_{r \to \infty} \limsup_{n \to \infty} P\{|\xi_n| > r\} = 0$$

because if ξ is a finite random variable,

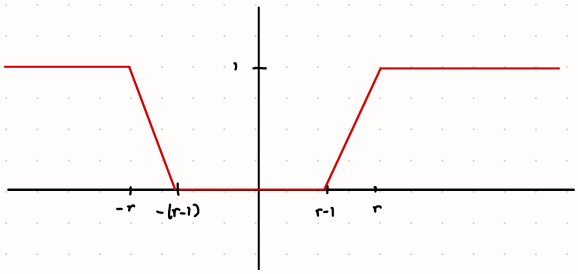
$$\lim_{r \to \infty} P\{|\xi| > r\} = 0$$

and this also holds for sups over finite collections.

Lemma 18.7: Weak Convergence and Tightness

Let $\xi, \xi_1, \xi_2, \ldots$ be random vectors in $\mathbb{R}^d$ such that $\xi_n \xrightarrow{d} \xi$. Then (ξ_n) is tight.

Proof: For any $r > 0$, define $f(x) = (1 - (r - |x|)_+)_+$. Then, this looks like



so

$$(1 - (r - |x|)_+)_+ = \begin{cases} 0 & 0 \le |x| \le r - 1 \\ \text{LERP} & r - 1 \le |x| \le r \\ 1 & r \le |x| \end{cases}$$

Thus,

$$\limsup_{n\to\infty} P\{|\xi_n| > r\} \le \lim_{n\to\infty} \mathbb{E}f(\xi_n) = \mathbb{E}f(\xi) \le P\{|\xi| > r - 1\}$$

so as the RHS tends to 0, we see that (ξ_n) is tight.

■

Lemma 18.8: Tightness and Convergence in Probability

Let $\xi_1, \xi_2, \ldots$ be random vectors in $\mathbb{R}^d$. Then (ξ_n) is tight iff $c_n\xi_n \xrightarrow{P} 0$ for any $c_1, c_2, \ldots \ge 0$ with $c_n \to 0$.

Proof: Assume that (ξ_n) is tight, and let $c_n \to 0$. Letting $r, \epsilon > 0$, and noting that $c_n r < \epsilon$ for all but finitely many $n \in \mathbb{N}$, we see that

$$\limsup_{n\to\infty} P\{|c_n\xi_n| > \epsilon\} \le \limsup_{n\to\infty} P\{|\xi_n| > r\}$$

Now the right hand side tends to zero as $r \to \infty$, so $P\{|c_n\xi_n| > \epsilon\} \to 0$. Thus, we see that $\xi_n \xrightarrow{P} 0$.

Now, if (ξ_n) is not tight, then we can find a sequence $(n_k) \subseteq \mathbb{N}$ such that $\inf_k P\{|\xi_{n_k}| > k\} > 0$. Thus, letting $c_n = \sup\{k^{-1} : n_k \ge n\}$, we see that $c_n \to 0$, but the subsequence $P\{|c_{n_k}\xi_{n_k}| > 1\} \not\to 0$, so $c_n\xi_n$ cannot converge in probability to 0.

■

Definition 18.9: Uniform Integrability

A family of random variables ξ_t, $t \in T$ is said to be *uniformly integrable* if

$$\lim_{r\to\infty} \sup_{t\in T} \mathbb{E}[|\xi_t| ; |\xi_t| > r] = 0$$

For (ξ_n) in L^1, this is equivalent to

$$\lim_{r\to\infty} \limsup_{n\to\infty} \mathbb{E}[|\xi_n| \,; |\xi_n| > r] = 0$$

as for $\xi \in L^1$, $\lim_{r\to\infty} \mathbb{E}[|\xi| \,; |\xi| > r] = 0$.

Note that the condition holds if the ξ_t are L^p-bounded for some $p > 1$ ($\sup_t \mathbb{E}\,|\xi_t|^p < \infty$) because

$$\mathbb{E}[|\xi_t| \,; |\xi_t| > r] \le r^{-p+1}\, \mathbb{E}\,|\xi_t|^p\,, \quad r > 0,\ p \ge 1$$

and $r^{-p+1} \to 0$ as $r \to \infty$ when $p > 1$.

11/5/26 Lecture 19: Uniform Integrability and Convergence of Means

We want to know when $\xi, \xi_1, \xi_2, \dots$ in L^1 with $\xi_n \xrightarrow{d} \xi$ implies $\mathbb{E}[\xi_n] \to \mathbb{E}[\xi]$.

Lemma 19.1: Uniform Integrability

Random variables ξ_t, $t \in T$ are uniformly integrable iff $\sup_t \mathbb{E}\,|\xi_t| < \infty$ and

$$\lim_{PA \to 0} \sup_{t \in T} \mathbb{E}[|\xi_t|\,; A] = 0$$

that is, for all $\epsilon > 0$, there exists $\delta > 0$ such that

$$P(A) < \delta \implies \mathbb{E}[|\xi_t|\,; A] < \epsilon \quad \forall t \in T$$

Proof: If ξ_t is uniformly integrable, then as

$$\mathbb{E}[|\xi_t|\,; A] \le rPA + \mathbb{E}[|\xi_t|\,; |\xi_t| > r], \quad r > 0$$

taking $PA \to 0$ and $r \to \infty$ at appropriate rates, we can get $\mathbb{E}[|\xi_t|\,; A] < \epsilon$ for all t, for any $\epsilon > 0$.

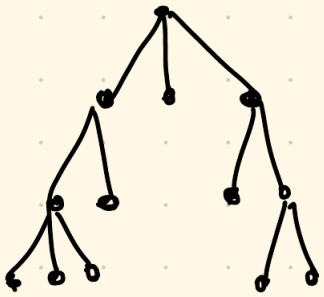
For boundedness, let $A = \Omega$ and take r sufficiently large.

Conversely, let ξ_t be L^1-bounded and satisfy the above condition. By Chebyshev,

$$\sup_t P\{|\xi_t| > r\} \le r^{-1} \sup_t \mathbb{E}\,|\xi_t| \to 0$$

as $r \to \infty$. Thus, if we set $A = \{|\xi_t| > r\}$, we see $PA \to 0$ uniformly as $r \to \infty$, so $\mathbb{E}[|\xi_t|\,; |\xi_t| > r] \to 0$. ■

Example: Critical Galton-Watson Branching Process



Clearly, $\mathbb{E}[\#\text{ offspring in } n^{\text{th}} \text{ generation}] = m^n$, where m is the expected number of offspring of a single person.

If $m > 1 \implies$ positive probability the tree never dies out.

If $m < 1 \implies$ dies out with probability 1.

If $m = 1 \implies$ dies out almost surely!

- $\mathbb{E}[Z_n] = 1$, but $\mathbb{E}[Z_n] \not\to 0 = \mathbb{E}[\lim_n Z_n]$

Lemma 19.2: Convergence of Means

Let $\xi, \xi_1, \xi_2, \ldots$ be $\mathbb{R}_+$-valued random variables with $\xi_n \xrightarrow{d} \xi$. Then,

(i) $\mathbb{E}\xi \le \liminf_n \mathbb{E}\xi_n$

(ii) $\mathbb{E}\xi_n \to \mathbb{E}\xi < \infty$ iff the ξ_n are uniformly integrable; that is,

$$\lim_{r\to\infty} \limsup_{n\to\infty} \mathbb{E}[|\xi_n| \,;\, |\xi_n| > r] = 0$$

Proof: **(i)** For any $r > 0$, $x \mapsto x \wedge r$ is bounded and continuous on $\mathbb{R}_+$, so

$$\liminf_{n\to\infty} \mathbb{E}[\xi_n] \ge \lim_{n\to\infty} \mathbb{E}[\xi_n \wedge r] = \mathbb{E}[\xi \wedge r]$$

as $\xi_n \to \xi$ in distribution. Letting $r \to \infty$, we see

$$\liminf_{n\to\infty} \mathbb{E}[\xi_n] \ge \mathbb{E}[\xi]$$

where we can turn $\liminf$ to $\lim$ as we know the limit exists.

(ii) ($\Leftarrow$) Assume uniform integrability. Then,

$$\mathbb{E}[\xi] \le \liminf_n \mathbb{E}[\xi_n] < \infty$$

(since they are L^1-bounded by above). Then, for any $r > 0$,

$$\begin{aligned}|\mathbb{E}[\xi_n] - \mathbb{E}[\xi]| \leq |\mathbb{E}[\xi_n] - \mathbb{E}[\xi_n \wedge r]| + |\mathbb{E}[\xi_n \wedge r] - \mathbb{E}[\xi \wedge r]| \\ + |\mathbb{E}[\xi \wedge r] - \mathbb{E}[\xi]|\end{aligned}$$

Now, $(x - x \wedge r) \leq x1\{x > r\}$, so as $\mathbb{E}[\xi_n; \xi_n > r]$ (uniform integrability) and $\mathbb{E}[\xi; \xi > r]$ (DCT) go to zero,

$$|\mathbb{E}[\xi_n] - \mathbb{E}[\xi_n \wedge r]| \to 0 \qquad |\mathbb{E}[\xi \wedge r] - \mathbb{E}[\xi]| \to 0$$

and by convergence in distribution,

$$|\mathbb{E}[\xi_n \wedge r] - \mathbb{E}[\xi \wedge r]| \to 0$$

so taking $n \to \infty$ and $r \to \infty$, we get $\mathbb{E}[\xi_n] \to \mathbb{E}[\xi]$.

($\Rightarrow$) Assume $\mathbb{E}\xi_n \to \mathbb{E}\xi < \infty$. Fix $r > 0$. Then, by continuity,

$$\mathbb{E}[\xi_n; \xi_n > r] \leq \mathbb{E}[\xi_n - \xi_n \wedge (r - \xi_n)_+] \to \mathbb{E}[\xi - \xi \wedge (r - \xi)_+]$$

since $x1\{x > r\} + x \wedge (r - x)_+ \leq x$. Then, as $x \wedge (r - x)_+ \uparrow x$ as $r \to \infty$, the RHS goes to 0 by DCT, so uniform integrability holds.

■